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On Strong Solvability of One Nonlocal Boundary Value Problem for Laplace Equation in Grand Sobolev Space in Rectangle

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Abstract. We consider a nonlocal boundary value problem for the Laplace equation in a rectangular domain in Sobolev spaces generated by the norm of the grand Lebesgue space. The concept of strong solvability of this problem is introduced and it's correct solvability is proved. At the same time, the basis property of the eigen and associated functions of one spectral problem in separable grand Lebesgue spaces is proved, and this fact is used to establish correct solvability. Note that earlier this problem in a semi-infinite strip in the classical formulation was considered in the works of E.I. Moiseev [24], M.E. Lerner and O.A. Repin [20].

Key Words and Phrases: Laplace equation, nonlocal problem, grand Lebesgue space, strong solvability.

2010 Mathematics Subject Classifications: 35A01; 35J05; 35K05

1. Introduction

The theory of the strong and weak solvability of linear elliptic equations in the Sobolev spaces is well developed and can be found in the classical monographs. In spite of this, a lot of problems, arising in the mechanics and the mathematical physics do not fit to this theory. An example, of such a problem is the following degenerate elliptic equation, studied by Moiseev in [24].

Consider the following (formal for now) nonlocal boundary value problem for the Laplace equation:

$$u_{xx} + u_{yy} = 0, \quad 0 < x < 2\pi, \quad 0 < y < h, \tag{1}$$

$$u(x,0) = \varphi(x), u(x,h) = \psi(x), \qquad 0 < x < 2\pi, \tag{2}$$

$$u_x(0,y) = 0, u(0,y) = u(2\pi,y), \ 0 < y < h. \tag{3}$$

Such problems have specific peculiarities compared to the ones with local conditions. Earlier, F.I.Frankl [13]; [14, p. 453-456] considered the problem with nonlocal boundary condition for a mixed type equation. Bitsadze-Samarski problem [10] for elliptic equations

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is also nonlocal with supports on a part of the boundary of domain, and these supports are free of other boundary conditions. In [18], N.I.Ionkin and E.I.Moiseev solved the boundary value problem for multi-dimensional parabolic equations with nonlocal conditions, whose supports are the characteristic and the improper parts of the boundary of domain.

We note that recently, interest in nonstandard functions spaces has greatly increased in connection with their applications in mechanics, mathematical physics and pure mathematical problems. Such spaces include Lebesgue spaces with a variable summability exponent, Morrey spaces, grand Lebesgue spaces, Orlicz, Lorents, Martsinkevich, etc.. That's why the number of research works dedicated has been growing in recent years (see, e.g., [2, 3, 5, 6, 7, 8, 9, 11, 12, 21, 25]), and the elaboration of a corresponding theory is far from complete. This article is also devoted to this direction. Let us note, that this problem cannot be treated with the classical methods developed for linear elliptic operators. Our method is based on spectral theory, which is used, for example, in the works [19, 21, 24].

2. Auxiliary concepts and facts

We will use standard notations. N will be the set of positive integers, while $\alpha = (\alpha_1;\alpha_2) \in Z^+ \times Z^+$ will denote a multi-index, where $Z^+ = N \cup \{0\}$. Denote $\partial^\alpha u = \frac{\partial^{|\alpha|}u}{\partial x^{\alpha_1}\partial y^{\alpha_2}}$, where $|\alpha| = \alpha_1 + \alpha_2$. By $|M|$ we will denote the Lebesgue measure of the set M; $\overline{M}$ will be the closure of M. $C^\infty\left(\overline{M}\right)$ will stand for the infinitely differentiable functions on $\overline{M}$, and $C_0^\infty(M)$ will denote the infinitely differentiable and finite functions on M. Throughout this paper we will assume that p' is a conjugate number of $p, 1 < p < +\infty$: $\frac{1}{p'} + \frac{1}{p} = 1$. $d\sigma$ is an area element. We also accept $p_\varepsilon = p - \varepsilon$.

Let us define our weighted grand Sobolev space. Let, $\Pi = (0, 2\pi) \times (0, h)$. Denote by $L_{p)}(\Pi)$ a Banach space of functions on Π with the mixed norm

$$\|f\|_{L_{p)}(\Pi)} = \sup_{0<\varepsilon<p-1} \int_0^h \left(\varepsilon \int_0^{2\pi} |f(x;y)|^{p-\varepsilon} dx\right)^{\frac{1}{p-\varepsilon}} dy, 1 < p < +\infty.$$

Denote by $W^2_{p)}(\Pi)$ a grand Sobolev space generated by the norm

$$\|u\|_{W^2_{p)}(\Pi)} = \sum_{|\alpha|\le 2} \|\partial^\alpha u\|_{L_{p)}(\Pi)}.$$

Now denote by $L_{p)}(I)$, where $I = (0, 2\pi)$, a grand Lebesgue space generated by the norm

$$\|f\|_{L_{p)}(I)} = \sup_{0<\varepsilon<p-1} \left(\varepsilon \int_I |f(x)|^{p-\varepsilon} dx\right)^{\frac{1}{p-\varepsilon}}.$$

We will also consider the grand Sobolev space $W^2_{p)}(I)$, generated by the norm

$$\|f\|_{W^2_{p)}(I)} = \|f\|_{L_{p)}(I)} + \|f'\|_{L_{p)}(I)} + \|f''\|_{L_{p)}(I)}.$$

These spaces are non-separable and therefore the method of biorthogonal expansion (essentially the spectral method) is not applicable for studying the solvability of differential equations with respect to these spaces. In this regard we select the subspace $N_{p)}(\Pi)\subset L_{p)}(\Pi)$ (separable) based on the shift operator T_δ:

$$(T_\delta u)(x;y) = \begin{cases} u(x+\delta:y) & (x+\delta:y)\in\Pi, \\ 0 & (x+\delta:y)\notin\Pi. \end{cases}$$

So let us assume

$$N^2_{p)}(\Pi) = \left\{ W^2_{p)}(\Pi) : \sum_{|\alpha|\le 2} \|T_\delta(\partial^\alpha u) - \partial^\alpha u\|_{L_{p)}(\Pi)} \to 0, \delta\to 0 \right\}.$$

Definition 1. *A system $\{u_n\}_{n\in N}\subset X$ is called a basis if any element $f\in X$ is uniquely represented as a series*

$$f = \sum_{n=1}^{\infty} c_n u_n$$

convergent in the norm X.

Definition 2. *A system $\{u_n\}_{n\in N}\subset X$ is called complete in X if $\overline{Sp}\{u_n\} = X$ and minimal in X if $u_n\notin \overline{Sp}\{u_k\}_{k\ne n}$.*

It is known that each basis of the space X is a complete and minimal system in X, the converse is not true in general.

Minimum criterion. The system $\{u_N\}_{n\in N}$ is minimal in X if and only if there exists a biorthogonal system i.e. there exists a system $\{\vartheta_n\}_{n\in N}\subset X^*$ such that $\langle u_n,\vartheta_k\rangle = \vartheta_k(u_n) = \delta_{nk}$, where δ_{nk} is the Kronecker symbol.

Basis criterion. The system $\{u_n\}_{n\in N}\subset X$ is a basis of the space X if and only if the following conditions are satisfied.

1. $\{u_n\}_{n\in N}$ is complete and minimal in X;
2. *uniformly bounded projectors*

$$P_n f = \sum_{k=1}^{n} \langle f,\vartheta_k\rangle u_k,$$

where $\{v_k\}_{k\in N}$ is a biorthogonal system.

Definition 3. *A system $\{u_n\}_{n\in N}\subset X$ is called a basis with brackets in X if there exists a sequence of integers $0=n_0<n_1<n_2<\ldots$, such that each element of $f\in X$ is uniquely represented as a series*

$$f = \sum_{k=0}^{\infty}\sum_{i=n_k+1}^{n_{k+1}} c_i u_i,$$

convergent in the norm X.

To obtain our main results we will extensively use also the following Minkowski's inequality for integrals.

Proposition 1 (Minkowski's inequality). *Let $(M_k;\sigma_{M_k};\mu_k)$, $k=\overline{1,2}$, be measurable spaces with σ-finite measures μ_k and $F(x;y)$ be a $\mu_1\times\mu_2$-measurable function. Then*

$$\left\|\int_{M_1} F(x;y)\,d\mu_1(x)\right\|_{L_{p)}(\mu_2)} \leq \int_{M_1} \|F(x;\cdot)\|_{L_{p)}(\mu_2)} d\mu_1(x),$$

where

$$\|f\|_{L_{p)}(\mu)} = \sup_{0<\varepsilon<p-1}\left(\varepsilon\int |f|^{p-\varepsilon}d\mu\right)^{\frac{1}{p-\varepsilon}}.$$

So let's introduce the following

Definition 4. *A function $u\in N^2_{p)}(\Pi)$ is called a strong solution of the problem (1)-(3) if the equality (1) is satisfied for a.e. $(x;y)\in\Pi$ and its trace $u|_{\partial\Pi}$ satisfies the relations (2), (3).*

Introduce the systems of functions $\{u_n(x)\}_{n\in Z^+}$ and $\{\vartheta_n(x)\}_{n\in Z^+}$, where

$$u_{2n}(x) = cos\,nx\ , n\in Z^+, \qquad u_{2n-1}(x) = x sin\,nx, n\in N, \tag{4}$$

$$\vartheta_0(x) = \frac{1}{2\pi^2}(2\pi - x), \vartheta_{2n}(x) = \frac{1}{\pi^2}(2\pi - x)\,cos\,nx\ , \vartheta_{2n-1}(x) = \frac{1}{\pi^2} sin\,nx\ , n\in N. \tag{5}$$

Note that these systems are biorthogonal conjugate, which can be verified directly. To obtain our main result, we will significantly use the following theorem.

Theorem 1. *The system (4) forms a basis for $N_{p)}(I)$.*

Proof. Conjugate space of $L_{p)}(I)$ is $L_{p')}(I)$. It is absolutely clear that the system (4) belongs to $L_{p')}(I)$ and is biorthonormalized to the system (4) (see [1]). It follows that (4) is minimal in $L_{p)}(I)$. On the other hand, from [1] it follows that the system (4) forms a basis with brackets for $N_{p)}(I)$ for every $p\in(1,+\infty)$, and, consequently, it is complete in $L_{p_1)}(I)$. Then from the embedding $N_{p_1)}(I)\subset L_{p)}(I)$ it follows that (4) is complete and, consequently, complete and minimal in $L_{p)}(I)$.

Let's prove the basicity of the system (4) for $N_{p)}(I)$. Consider the projectors

$$P_n(f) = \sum_{k=0}^{n}\langle f,\vartheta_k\rangle u_k, \forall n\in Z^+, \forall f\in N_{p)}(I)$$

where

$$\langle f,g\rangle = \int_0^{2\pi} f(x)\overline{g(x)}dx.$$

From the basicity with brackets of the system (4) for $N_{p)}(I)$ it follows that

$$\exists C>0 : \|P_{2n}(f)\|_{L_{p)}(I)} \leq C\|f\|_{L_{p)}(I)}, \forall n\in N. \tag{6}$$

On the other hand, from (4), (6) we have

$$\exists M > 0 : \|u_n\|_{L_{p)}(I)} \le M, \qquad \|\vartheta_n\|_{L_{p')}} \le M, \qquad \forall n \in N. \tag{7}$$

Considering the relations (6), (7), we obtain

$$\|P_{2n+1}(f)\|_{L_{p)}(I)} = \|P_{2n}(f)\,\langle f, \vartheta_{2n+1}\rangle\, u_{2n+1}\|_{L_{p)}(I)} \le \|P_{2n}\,(f)\|_{L_{p)}(I)} +$$
$$+\|\langle f, \vartheta_{2n+1}\rangle\, u_{2n+1}\|_{L_{p)}(I)} \le C\|f\|_{L_{p)}(I)} + \|f\|_{L_{p)}(I)}\|u_n\|_{L_p(I)}\|\vartheta_n\|_{L_{p')}(I)} \le$$
$$\le \left(C + M^2\right)\|f\|_{L_{p)}(I)}. \tag{8}$$

From (6), (8) it follows that the projectors $\{P_n\}_{n\in Z^+}$ are uniformly bounded, and, according to the criterion for basicity, this means that the system (4) forms a basis for $N_{p)}\,(I)$. The theorem is proved.

3. Main Results

In this section, we will study the existence and uniqueness of strong solution of the problem (1)-(3) in the sense of Definition 4. First, denote $\Gamma_0 = \{(0; y) : 0 < y < h\}$ and $\Gamma_{2\pi} = \{(2\pi; y)\,, 0 < y < h\}$. Consider the following nonlocal problem:

$$\Delta u = 0,\ (x; y) \in \Pi, \tag{9}$$

$$u|_{I_0} = \varphi,\ u|_{I_h} = \psi,\ u|_{\Gamma_0} = u|_{\Gamma_{2\pi}},\ u_x|_{\Gamma_0} = 0. \tag{10}$$

By the solution of this problem, we mean a function $u \in N^2_{p)}\,(\Pi)$, which satisfies the equality (9) a.e. in Π and whose traces satisfy the relations (10) on the boundary $\partial\Pi = I_0 \cup I_h \cup \Gamma_0 \cup \Gamma_{2\pi}$. Let's first prove the uniqueness of the solution. The following theorem is true:

Theorem 2. *The functions $\varphi, \psi \in N^2_{p)}\,(I)$ satisfy the conditions $\varphi\,(2\pi) - \varphi\,(0) = \varphi'\,(0) = 0, \psi\,(2\pi) - \psi\,(0) = \psi'\,(0) = 0$. If the problem (1)-(3) has a solution in $N^2_{p)}\,(\Pi)$, then it is unique.*

Proof. Suppose $u\,(x, y) \in N^2_{p)}\,(\Pi)$ is a solution of the problem (1)-(3). Consider $u_n\,(y) = \langle u\,(\cdot, y)\,, \vartheta_n\,(\cdot)\rangle$, i.e.

$$\left.\begin{array}{r} U_0\,(y) = \frac{1}{2\pi^2}\int_0^{2\pi} u\,(x, y)\,(2\pi - x)dx \\ U_{2n}\,(y) = \frac{1}{\pi^2}\int_0^{2\pi} u\,(x, y)\,(2\pi - x)cos\,nx\;dx \quad , \\ U_{2n-1}\,(y) = \frac{1}{\pi^2}\int_0^{2\pi} u\,(x, y)sin\,nx \quad dx, n \in N. \end{array}\right\} \tag{11}$$

we obtain the following relations for $U_{2n-1}\,(y)$ (respectively, for $U_{2n}\,(y)$):

$$U''_{2n-1}\,(y) - n^2 U_{2n-1}\,(y) = 0, y \in (0.h)\,, \tag{12}$$

$$U_{2n}''(y) - n^2 U_{2n}(y) = -2n U_{2n-1}(y), y \in (0.h). \tag{13}$$

By Newton-Leibniz formula we have

$$u(x,\xi) = u(x,0) + \int_0^\xi \frac{\partial u(x,y)}{\partial y} dy = \varphi(x) + \int_0^\xi \frac{\partial u(x,y)}{\partial \xi} d\xi, \quad \text{a.e. } x \in I.$$

Consequently,

$$|u(x,\xi) - \varphi(x)| \leq \int_0^\xi \left| \frac{\partial u(x,y)}{\partial y} \right| dy, \quad \text{a.e.} \quad x \in I.$$

Hence it immediately follows that

$$\int_I |u(x,\xi) - \varphi(x)| dx \leq \int_I \int_0^\xi \left| \frac{\partial u(x,y)}{\partial y} \right| dy dx. \tag{14}$$

We have $|\Pi_\xi| \to 0$ as $\xi \to +0$. Then from (14) it follows that

$$u_\xi(\cdot) \to \varphi(\cdot), \xi \to +0, \tag{15}$$

in the norm of the space $L_1(I)$.
Similarly we have

$$u(x,\xi) = u(x,h) - \int_\xi^h \frac{\partial u(x,y)}{\partial y} dy = \psi(x) - \int_\xi^h \frac{\partial u(x,y)}{\partial y} dy, \quad \text{a.e. } x \in I.$$

Hence,

$$\int_I |u(x,\xi) - \psi(x)| dx \leq \int_I \int_\xi^h \left| \frac{\partial u(x,y)}{\partial y} \right| dy dx. \tag{16}$$

As $|\Pi \backslash \Pi_\xi| \to 0$ when $\xi \to h-0$, from (16) it follows that

$$u_\xi(\cdot) \to \psi(\cdot), \quad \xi \to h-0, \tag{17}$$

in the norm of the space $L_1(I)$.

On the other hand, it is clear that $U_n(y) \in W_1^2(0,h)$. Hence it immediately follows that there exist the limits

$$\lim_{y \to +0} U_n(y) = U_n(0), \lim_{y \to h-0} U_n(y) = U_n(h), \forall n \in Z^+.$$

By (15) and (17), from the last two relations it immediately follows that

$$U_n(0) = \varphi_n, \qquad U_n(h) = \psi_n, \qquad \forall n \in Z^+, \tag{18}$$

where

$$\begin{cases} \varphi_0 = \frac{1}{2\pi^2}\int_0^{2\pi}\varphi(x)(2\pi - x)\,dx, \\ \varphi_{2n-1} = \frac{1}{\pi^2}\int_0^{2\pi}\varphi(x)\,sin\,nx\;dx, \\ \varphi_{2n} = \frac{1}{\pi^2}\int_0^{2\pi}\varphi(x)(2\pi - x)\,cos\,nx\;dx, \qquad n \in N; \end{cases} \tag{19}$$

$$\begin{cases} \psi_0 = \frac{1}{2\pi^2}\int_0^{2\pi}\psi(x)(2\pi - x)\,dx, \\ \psi_{2n-1} = \frac{1}{\pi^2}\int_0^{2\pi}\psi(x)\,sin\,nx\;dx, \\ \psi_{2n} = \frac{1}{\pi^2}\int_0^{2\pi}\psi(x)(2\pi - x)\,cos\,nx\;dx, \qquad n \in N. \end{cases} \tag{20}$$

The solution of the problem (12), (18) is

$$U_{2n-1}(y) = \psi_{2n-1}\frac{sinh\,ny}{sinh\,nh} + \varphi_{2n-1}\frac{sinh\,n(h-y)}{sinh\,nh}, \forall n \in N, \tag{21}$$

and the solution of the problem (13), (18) is

$$U_0(y) = \frac{\psi_0 - \varphi_0}{h}y + \varphi_0, \tag{22}$$

$$U_{2n}(y) = \frac{\psi_{2n}sin\,nh + h\psi_{2n-1}cos\,nh - h\varphi_{2n-1}}{sinh\,nh}\frac{sinh\,ny}{sinh\,nh} +$$
$$+\varphi_{2n}\frac{sinh\,n(h-y)}{sinh\,nh} - y(\psi_{2n-1}\frac{cosh\,ny}{sinh\,nh} - \varphi_{2n-1}\frac{cosh\,n(h-y)}{sin\,nh}, \forall n \in N. \tag{23}$$

Now we can proceed to the proof of the uniqueness of solution. For this, it suffices to prove that the corresponding homogeneous problem has only trivial solution. In fact, if $\varphi(x) = \psi(x) \equiv 0$, then $\varphi_n = \psi_n = 0, \forall n \in Z^+$, and from the formulas (21)-(23) it follows that $U_n(y) = 0$, $\forall y \in (0,h), \forall n \in Z^+$. As $u_y \in N_{p)}(I)$, $\forall y \in (0,h)$, the basicity of the system (4) for $N_{p)}(I)$ implies $u_y(x) = 0$ a.e. $x \in I$ and $\forall y \in (0,h)$. Hence it follows that $u(x;y) = 0$ a.e. $(x;y) \in \Pi$. Consequently, the homogeneous problem has only trivial solution, and this completes the proof of uniqueness.

Now let's prove the existence of solution. The following theorem is true:

Theorem 3. *Let the boundary functions $\varphi(x)$ and $\psi(x)$ belong to the space $N^2_{p)}(I)$ and satisfy the conditions*

$$\varphi(0) - \varphi(2\pi) = \varphi'(0) = 0, \psi(0) - \psi(2\pi) = \psi'(0) = 0.$$

Then the problem (1)-(3) *has a (unique) solution in $N^2_{p)}(\Pi)$.*

Proof. Consider the function

$$u(x,y) = U_0(y) + \sum_{n=1}^{\infty} U_n(y)\,u_n(x) = U_0(y) +$$

$$+\sum_{k=1}^{\infty}\left(U_{2k}\left(y\right)cos\,kx+U_{2k-1}\left(y\right)xsin\,kx\right),(x,y)\in\Pi, \tag{24}$$

where the coefficients $U_0\left(y\right), U_{2k}\left(\cdot\right), U_{2k-1}\left(\cdot\right), k\in N$, are defined by (21)-(23). Let's show that the function $u\left(x,y\right)$ belongs to $N^2_{p)}\left(\Pi\right)$. Denote by $u_{\alpha_1,\alpha_2}\left(x,y\right)$ the sum of the series obtained by the formal differentiation of the series (24), i.e.

$$u_{\alpha_1,\alpha_2}\left(x,y\right)=U_0^{(\alpha_2)}\left(y\right)+\sum_{\text{n}=1}^{\infty}U_n^{(\alpha_2)}\left(y\right)u_n^{(\alpha_1)}\left(x\right), \tag{25}$$

where $\alpha_1,\alpha_2\in Z^+,\alpha_1+\alpha_2=0,1,2;$ $u_{0,0}\left(x,y\right)=u\left(x,y\right)$ and $U_n^{(\alpha_2)}\left(y\right)=\frac{d^{\alpha_2}U_n}{dy^{\alpha_2}}$; $U_n^{(\alpha_1)}\left(x\right)=\frac{d^{\alpha_1}U_n}{dx^{\alpha_1}}$.

Let us first consider the following member of series (24).

$$u_1\left(x,y\right)=\sum_{k=1}^{\infty}U_{2k-1}\left(y\right)\ xsin\,kx\ .$$

So, differentiating this series formally term-by-term, we have

$$\frac{\partial^2u_1}{\partial y^2}=\sum_{k=1}^{\infty}U''_{2k-1}\left(y\right)xsin\,kx=\sum_{k=1}^{\infty}k^2U_{2k-1}\left(y\right)xsin\,kx, \tag{26}$$

$$\frac{\partial u_1}{\partial x}=\sum_{k=1}^{\infty}U_{2k-1}\left(y\right)sin\,kx\ +\sum_{k=1}^{\infty}kU_{2k-1}\left(y\right)xcos\,kx, \tag{27}$$

$$\frac{\partial^2u_1}{\partial x^2}=2\sum_{k=1}^{\infty}kU_{2k-1}\left(y\right)cos\,kx-\sum_{k=1}^{\infty}k^2U_{2k-1}\left(y\right)xsin\,kx. \tag{28}$$

Denote

$$w\left(x,y\right)=\sum_{k=1}^{\infty}k^2U_{2k-1}\left(y\right)x\,sin\,kx.$$

Let's show that the function $w\left(x,y\right)$ belongs to $N_{p)}\left(\Pi\right)$. Let

$$\varphi''_{2k-1}=\frac{1}{\pi^2}\int_0^{2\pi}\varphi''\left(x\right)sin\,kxdx,\ \ \psi''_{2k-1}=\frac{1}{\pi^2}\int_0^{2\pi}\psi''\left(x\right)sin\,kxdx.$$

From (19), integrating by parts, we obtain

$$\varphi_{2k-1}=-\frac{1}{\pi^2k}\int_0^{2\pi}\varphi\left(x\right)dcos\,kx\ =-\frac{1}{\pi^2k}\left(\varphi\left(2\pi\right)-\varphi\left(0\right)-\int_0^{2\pi}\varphi'\left(x\right)cos\,kx\ dx\right)=$$

$$=\frac{1}{\pi^2k}\int_0^{2\pi}\varphi'\left(x\right)cos\,kx\ dx\ =\frac{1}{\pi^2k^2}\int_0^{2\pi}\varphi''\left(x\right)sin\,kx\ dx\ =\frac{1}{k^2}\varphi''_{2k-1}.$$

Similarly, from (20) we have

$$\psi_{2k-1} = -\frac{1}{k^2}\psi''_{2k-1}.$$

Thus,

$$w(x,y) = \sum_{k=1}^{\infty}\left(\psi''_{2k-1}\frac{\sinh ky}{\sinh kh} + \varphi''_{2k-1}\frac{\sinh k(h-y)}{\sinh kh}\right) x \sin kx \ .$$

Let $\varepsilon \in (0, p-1)$ be an arbitrary number.We have the following continuous embeddings $L_p(I) \subset L_{p-\varepsilon}(I) \subset L_1(I)$. Let us suppose that $\beta = \frac{p}{p-\varepsilon} \Longrightarrow \frac{1}{\beta'} = 1 - \frac{p-\varepsilon}{p} = \frac{\varepsilon}{p}$. Applying the Holder's inequality, we have

$$\int_0^{2\pi} |f|^{p-\varepsilon} dx = \int_0^{2\pi} |f|^{p-\varepsilon} dx \leq \left(\int_0^{2\pi} |f|^p dx\right)^{\frac{1}{\beta}} \left(\int_0^{2\pi} dx\right)^{\frac{1}{\beta'}} \Longrightarrow$$

$$\Longrightarrow \left(\varepsilon \int_0^{2\pi} |f|^{p-\varepsilon} dx\right)^{\frac{1}{p-\varepsilon}} \leq \left(\int_0^{2\pi} |f|^p dx\right)^{\frac{1}{p}} \left(\int_0^{2\pi} dx\right)^{\frac{\varepsilon}{p-\varepsilon}\frac{1}{p}} \varepsilon^{\frac{1}{p-\varepsilon}} \leq c\left(\int_0^{2\pi} |f|^p dx\right)^{\frac{1}{p}},$$

where $c > 0$ is a constant which is independent of f and ε. This immediately follows

$$\|f\|_{L_{p)}(I)} \leq c\|f\|_{L_p(I)}, \forall f \in L_{p)}(I).$$

Let $\exists\ \alpha > 1$ and $\frac{1}{\alpha} + \frac{1}{\alpha'} = 1$.

Applying Holder's inequality again, we obtain

$$\left(\int_0^{2\pi} |w(x,y)|^p dx\right)^{\frac{1}{p}} \leq \left(\int_0^{2\pi} 1 dx\right)^{\frac{1}{\alpha}} \left(\int_0^{2\pi} |w(x,y)|^{p\alpha'} dx\right)^{\frac{1}{p\alpha'}} = c\left(\int_0^{2\pi} |w(x,y)|^{p_1} dx\right)^{\frac{1}{p_1}},$$

where $c = (2\pi)^{\frac{1}{\alpha}}$ (consequently, does not depend on $w(x,y)$) and $p_1 = p\alpha'$. Let's consider the cases $p \geq 2$ and $1 < p < 2$. Consider the following separate cases regarding p.

I. $p \geq 2$. From the previous inequality we have

$$\|w(.;y)\|_{L_{p)}} \leq c\left(\int_0^{2\pi} |w(x,y)|^p dx\right)^{\frac{1}{p}} \leq c\sum_{k=1}^{\infty} |U_{2k-1}(y)| \leq c_1 \sum_{k=1}^{\infty} |U_{2k-1}(y)| \leq$$

$$\leq c_1 \sum_{k=1}^{\infty} \left|\psi''_{2k-1}\frac{\sinh ky}{\sinh kh} + \varphi''_{2k-1}\frac{\sinh k(h-y)}{\sinh kh}\right| \leq$$

$$\leq c_1 \sum_{k=1}^{\infty} \left(\left|\psi''_{2k-1}\right| \frac{\sinh ky}{\sinh kh} + \left|\varphi''_{2k-1}\right| \frac{\sin k(h-y)}{\sinh kh}\right).$$

Hence, first integrating with respect to $y \in (0,h)$ and then applying Holder's inequality for any $\beta \in (1,\infty)$, we obtain

$$\|w\|_{L_{p)}(\Pi)} \leq c_1 \sum_{k=1}^{\infty} \left(\frac{\left|\psi''_{2k-1}\right|}{\sinh kh} \int_0^h \sinh ky dy + \frac{\left|\varphi''_{2k-1}\right|}{\sinh kh} \int_0^h \sinh k(h-y) dy\right) \leq$$

$$\leq c_1 \sum_{k=1}^{\infty} \frac{\left|\psi''_{2k-1}\right| + \left|\varphi''_{2k-1}\right|}{\sin kh} \int_0^h \sinh ky dy \leq$$

$$\leq c_1 \sum_{k=1}^{\infty} \frac{\cosh kh - 1}{k\sinh kh} \left(\left|\psi''_{2k-1}\right| + \left|\varphi''_{2k-1}\right| \right) \leq$$

$$\leq c_2 \sum_{k=1}^{\infty} \frac{1}{k} \left(\left|\psi''_{2k-1}\right| + \left|\varphi''_{2k-1}\right| \right) \leq c_2 \left(\sum_{k=1}^{\infty} \frac{1}{k^{\beta'}} \right)^{\frac{1}{\beta'}} \left(\left(\sum_{n=1}^{\infty} \left|\varphi''_n\right|^{\beta} \right)^{\frac{1}{\beta}} + \left(\sum_{n=1}^{\infty} \left|\psi''_n\right|^{\beta} \right)^{\frac{1}{\beta}} \right).$$

Now, assuming $\beta \geq 2$ and applying classical Hausdorff-Young inequality (see, e.g. [27, p. 154]. We have

$$\|w\|_{L_{p)}(\Pi)} \leq c_3 \left(\left\|\psi''\right\|_{L_{\beta'}(I)} + \left\|\varphi''\right\|_{L_{\beta'}(I)} \right). \tag{29}$$

Let us suppose that $r = \frac{p_\varepsilon}{q}$ and $g \in L_{p)}(I)$. Then $1 < r < p_\varepsilon$ and we have

$$\left(\int_I |g|^r dx \right)^{\frac{1}{r}} = \left(\int_I |g|^{\frac{p_\varepsilon}{q}} dx \right)^{\frac{1}{r}} \leq \left(\int_I |g|^{p_\varepsilon} dx \right)^{\frac{1}{qr}} \left(\left(\int_I dx \right) \right)^{\frac{1}{q' r}} =$$

$$= c \left(\int_I |g|^{p_\varepsilon} dx \right)^{\frac{1}{p_\varepsilon}}.$$

Then, the last inequality means $g \in L_r(I)$ and

$$\|g\|_{L_r(I)} \leq c\|g\|_{L_{p_\varepsilon}(I)}, \tag{30}$$

where $c > 0$ is a constant independent of g. Also note that the continuous embedding $L_{p)}(I) \subset L_\alpha(I)$ is true for every $\alpha \in (1, r)$. Let us choose β big enough to satisfy the condition

$$1 < \beta' < r \Rightarrow \|g\|_{L_{\beta'}(I)} \leq c\|g\|_{L_r(I)}$$

is satisfied. Then from inequalities (29),(30) we obtain

$$\|w\|_{L_{p)}(\Pi)} \leq c \left(\left\|\psi''\right\|_{L_{p)}(I)} + \left\|\varphi''\right\|_{L_{p)}(I)} \right).$$

II. $p \in (1, 2)$. Therefore, choosing $\alpha > 1$ close enough to 1, we can provide that $p_1 = p\alpha' > 2$ (this is possible, because $\alpha' \to +\infty$ as $\alpha \to 1 + 0$). With this, further considerations are carried out similar to the previous case.

Other series from (26)-(28), and, consequently, all series from (25) are estimated in a similar way. So, as a result, we obtain

$$\|u\|_{W^2_{p)}(\Pi)} \leq c \left(\|\varphi\|_{W^2_{p)}(I)} + \|\psi\|_{W^2_{p)}(I)} \right),$$

where $c > 0$ is a constant independent of φ and ψ. The fulfillment of equation (9) by $u(\cdot;\cdot)$ can be verified directly. Let's verify the fulfillment of boundary conditions. Denote the trace operators on $\Gamma_0, \Gamma_{2\pi}$, I_0 and I_h by $\theta_0, \theta_{2\pi}$, T_0 and T_h, respectively. Let's show that $T_0 u = \varphi$. It is clear that, $T_0 u \in L_1(I)$ *and* $\varphi \in L_1(I)$. From the boundedness of the operator $T_0 \in \left[W_p^2(\Pi); L_p(I)\right]$, $\forall\ p \geq 1$, it follows that if $u_m \to u$ in $W_{p)}^2(\Pi)$, then $u_m/_I \to u/_I$ in $L_{p)}(I)$.

Now, let's consider the following functions:

$$u_m(x,y) = U_0(y) + \sum_{n=1}^{m} \left(U_{2n}(y)\, cos\, nx\ + U_{2n-1}(y)\, x sin\, nx\ \right), (x;y) \in \Pi, m \in N.$$

We have

$$T_0 u_m = u_m(x,0) = U_0(0) + \sum_{n=1}^{m} \left(U_{2n}(0)\, cos\, nx\ + U_{2n-1}(0)\, x sin\, nx\ \right) =$$

$$= \frac{1}{2\pi^2} \int_0^{2\pi} \varphi(\tau)(2\pi - \tau)\, d\tau +$$

$$+ \sum_{n=1}^{m} \left(\frac{1}{\pi^2} \int_0^{2\pi} \varphi(\tau)(2\pi - \tau)\, cos\, n\tau d\tau cos\, nx\ + \frac{1}{\pi^2} \int_0^{2\pi} \varphi(\tau)\, sin\, n\tau d\tau sin\, nx \right). \quad (31)$$

It is clear that, $T_0 u_m \to T_0 u$. On the other hand, the basicity of the system (4) for $N_{p)}(I)$ implies $T_0 u_m \to \varphi, m \to \infty$, in $L_{p)}(I)$. Consequently, $T_0 u = \varphi$, a.e. on I.

Absolutely similar we can show that $T_h u_m \to \psi, m \to \infty$, in $L_{p)}(I)$. Consequently, $T_h u = \psi$, a.e. on I.

Consider the operators θ_0 and $\theta_{2\pi}$. It is clear that $\theta_0 u_m = \theta_{2\pi} u_m$, $\forall m \in N$. Obviously, $\theta_0 u_m \to \theta_0 u$ and $\theta_{2\pi} u_m = \theta_{2\pi} u \Rightarrow \theta_0 u = \theta_{2\pi} u$. Thus, the boundary conditions (10) are fulfilled.

The theorem is proved.

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Approximation of the Hilbert transform in Orlicz spaces

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Abstract. The Hilbert transform is the main part of the singular integral equations on the real line. Therefore, approximations of the Hilbert transform are of great interest. This article is devoted to the approximation of the Hilbert transform in Orlicz spaces by operators which introduced by V.R.Kress and E.Mortensen to approximate the Hilbert transform of analytic functions in a strip.

Key Words and Phrases: Hilbert transform, singular integral, approximation, Orlicz spaces.
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1. Introduction

The Hilbert transform of a function $u \in L_p(\mathbb{R})$, $1 \le p < \infty$ is defined as the Cauchy principle value integral [23]

$$(Hu)(t) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{u(\tau)}{t-\tau} d\tau, \quad t \in \mathbb{R},$$

where the integral is understood in the Cauchy principal value sense. It is well known (see [18, 23]) that the Hilbert transform of the function $u \in L_p(\mathbb{R})$, $1 \le p < \infty$, exists for almost all values of $t \in \mathbb{R}$. In case $1 < p < \infty$, the Hilbert transform is a bounded map in the space $L_p(\mathbb{R})$ and satisfies the equation:

$$H^2 = -I.$$

The Hilbert transform plays an important role in the theory and practice of signal processing operations in continuous system theory because of its relevance to such problems as envelope detection and demodulation, as well as its use in relating the real and imaginary components, and the magnitude and phase components of spectra. The Hilbert transform is the main part of the singular integral equations on the real line (see [32]). Therefore, approximations of the Hilbert transform are of great interest.

Many papers have dealt with the numerical approximation of the Hilbert Transform in the case of bounded intervals and the reader can refer to [2, 7**?** , 12, 13, 15, 16, 19, 20, 21, 28, 30, 33, 34, 36, 37, 38, 42, 43] and the references given there. On the other hand, the literature concerning the numerical integration on unbounded intervals is by far poorer than the one on bounded intervals. The case of the Hilbert Transform has been considered very little and the reader can consult [5, 11, 14, 15, 25, 27, 31, 39, 40, 41, 44]. In particular, in [25] the authors assume that the function u is analytic in the strip $\{z \in \mathbb{C} : |\Im z| < d\}$, in which case they show that the series $\frac{2}{\pi}\sum_{k\in\mathbb{Z},k\neq even} \frac{u(t+k\delta)}{-k}$ uniformly converges to $(Hu)(t)$ as $\delta \to 0$. In [9] the author replaces the above series with the following one $\frac{1}{\pi}\sum_{k\in\mathbb{Z}} \frac{u(t+(k+1/2)\delta)}{-k-1/2}$ for a suitable choice of the step $\delta \to 0$.

This article is devoted to the approximation of the Hilbert transform of functions from Orlicz spaces by operators of the form

$$(H_\delta u)(t) = \frac{1}{\pi}\sum_{k\in\mathbb{Z}} \frac{u(t+(k+1/2)\delta)}{-k-1/2}, \quad \delta > 0$$

which were introduced in [25]. It is proved that the operators are uniformly bounded maps in the Orlicz spaces, satisfies the equality

$$H_\delta^2 = -I,$$

and for any $\delta > 0$ the sequence of operators H_δ strongly converges to the operator H in these spaces. Note that for Lebesgue and Hölder spaces these approximations were considered respectively in the works [3, 4].

2. Orlicz spaces

Definition 1. *A convex left continuous function $\Phi : [0,\infty) \to [0,\infty]$ satisfying the conditions*

$$\lim_{r\to 0+} \Phi(r) = \Phi(0) = 0, \quad \lim_{r\to\infty} \Phi(r) = \infty$$

is called a Young function.

The class of Young functions satisfying the condition $\Phi(r) \in (0,\infty)$ for any $r \in (0,\infty)$ is denoted by $\mathcal{Y}$. If $\Phi \in \mathcal{Y}$, then Φ is absolutely continuous on each bounded closed interval and bijective from $[0,\infty)$ to $[0,\infty)$.

For the Young function Φ we will write

$$\Phi^{-1}(s) = \inf\{r \geq 0 : \ \Phi(r) > s\}, \quad 0 \leq s \leq \infty.$$

We note that if $\Phi \in \mathcal{Y}$, then Φ^{-1} is the usual inverse of the function Φ. And for each Young function, we have the inequalities

$$\Phi\left(\Phi^{-1}(r)\right) \leq r \leq \Phi^{-1}\left(\Phi(r)\right), \quad r \geq 0.$$

Definition 2. *If there exists a number $C > 1$ such that the inequality*

$$\Phi(2r) \leq C\Phi(r)$$

holds for any $r > 0$, then the Young function Φ is said to satisfy the Δ_2-condition (the notation: $\Phi \in \Delta_2$).

Definition 3. *If there exists a number $C > 1$ such that the inequality*

$$\Phi(r) \leq \frac{1}{2C}\Phi(Cr)$$

holds for any $r > 0$, then the Young function Φ is said to satisfy the ∇_2-condition (the notation: $\Phi \in \nabla_2$).

For the Young function Φ, the function $\tilde{\Phi}$ defined by the relation

$$\tilde{\Phi}(r) = \sup\{rs - \Phi(s) : s \in [0.\infty)\}, \quad r \geq 0,$$

is called the complementary function. The complementary function $\tilde{\Phi}$ also is a Young function and $\tilde{\tilde{\Phi}} = \Phi$. We note that $\Phi \in \Delta_2$ (accordingly, $\Phi \in \nabla_2$) if and only if $\tilde{\Phi} \in \nabla_2$ (accordingly, $\tilde{\Phi} \in \Delta_2$). For any $r > 0$, the following inequalities hold:

$$r \leq \Phi^{-1}(r)\,\tilde{\Phi}^{-1}(r) \leq 2r.$$

Definition 4. *Let Φ be a Young function. The set of measurable functions $f : \Omega \to \mathbb{R}$, $\Omega \subset \mathbb{R}$ satisfying the condition*

$$\exists k > 0 : \int\limits_{\Omega} \Phi(k\,|f(x)|)\,dx < \infty$$

is called the Orlicz space (the notation: $L_\Phi(\Omega)$).

The space $L_\Phi(\Omega)$ with norm

$$\|f\|_{L_\Phi} = \inf\left\{\lambda > 0 : \int\limits_{\Omega} \Phi\left(\frac{|f(x)|}{\lambda}\right) dx \leq 1\right\}$$

is a Banach space.

We note that if $\Phi(r) = r^p$, $1 \leq p < \infty$, then $L_\Phi(\Omega) = L_p(\Omega)$. The properties of Orlicz spaces are investigated in the works [1, 22, 24].

The following analogue of the Hölder's inequality is well known (see: [22]).

Theorem 1. *Let $\Omega \subset \mathbb{R}$ be a measurable set and functions f and g measurable on Ω. For a Young function Φ and its complementary function $\widetilde{\Phi}$, the following inequality is valid*

$$\int_{\Omega} |f(x)g(x)|dx \leq 2\|f\|_{L_\Phi(\Omega)}\|g\|_{L_{\widetilde{\Phi}(\Omega)}}.$$

Definition 5. *Let Φ be a Young function. The set of sequences $b = \{b_n\}_{n\in\mathbb{Z}}$ satisfying the condition*

$$\exists k > 0 : \sum_{n\in\mathbb{Z}} \Phi(k|b_n|) < \infty$$

is called the Orlicz sequence space (the notation: l_Φ).

The space l_Φ with norm

$$\|b\|_{l_\Phi} = \inf\left\{\lambda > 0 : \sum_{n\in\mathbb{Z}} \Phi\left(\frac{|b_n|}{\lambda}\right) \le 1\right\}$$

is a Banach space.

We note that if $\Phi(r) = r^p$, $1 \le p < \infty$, then $l_\Phi = l_p$. The properties of Orlicz sequence spaces are investigated in the works [17, 26, 29].

3. Properties of the approximating operators H_δ

The sequence $h(b) = \{(h(b))_n\}_{n\in\mathbb{Z}}$ is called the discrete Hilbert transform of the sequence $b = \{b_n\}_{n\in\mathbb{Z}}$, where

$$(h(b))_n = \sum_{m\neq n} \frac{b_m}{n-m}, \quad n \in \mathbb{Z}.$$

M.Riesz (see [35]) proved that if $b \in l_p$, $1 < p < \infty$, then $h(b) \in l_p$ and the inequality

$$\|h(b)\|_{l_p} \le C_p \|b\|_{l_p} \tag{1}$$

holds, where C_p is constant depending only on p.

We will use a modified version of the discrete Hilbert transform:

$$(\tilde{h}(b))_n = \sum_{m\in\mathbb{Z}} \frac{b_m}{n-m-1/2}, \quad n \in \mathbb{Z}.$$

K. Andersen [8] proved that the inequality (1) is also valid for the transform $\tilde{h}$, that is, there exist $\widetilde{C}_p > 0$ such that the inequality

$$\|\tilde{h}(b)\|_{l_p} \le \widetilde{C}_p \|b\|_{l_p} \tag{2}$$

holds for any $b \in l_p$, $1 < p < \infty$. Then it follows from Marcinkiewicz interpolation theorem in Orlicz spaces (see: [22]) that if $\Phi \in \Delta_2 \cap \nabla_2$, then the inequality (2) is also valid for the space l_Φ, that is, there exist $\widetilde{C}_\Phi > 0$ such that the inequality

$$\|\tilde{h}(b)\|_{l_\Phi} \le \widetilde{C}_p \|b\|_{l_\Phi}$$

holds for any $b \in l_\Phi$.

In [3], the authors prove that the operators H_δ are uniformly bounded maps in the spaces $L_p(\mathbb{R})$, $1 < p < \infty$ and

$$\|H_\delta\|_{L_p(\mathbb{R})\to L_p(\mathbb{R})} \le \|\tilde{h}\|_{l_p\to l_p}.$$

Then it follows from Marcinkiewicz interpolation theorem in Orlicz spaces (see: [22]) that if $\Phi \in \Delta_2 \cap \nabla_2$, then the operators H_δ are also uniformly bounded maps in the space $L_\Phi(\mathbb{R})$, that is there exist $C_\Phi > 0$ such that for any $\delta > 0$

$$\|H_\delta\|_{L_\Phi(\mathbb{R})\to L_\Phi(\mathbb{R})} \le C_\Phi. \tag{3}$$

Theorem 2. *For any $\delta > 0$ and $u \in L_\Phi(\mathbb{R})$ the following inequality holds:*

$$H_\delta(H_\delta u)(t) = -u(t). \tag{4}$$

Proof. For any $u \in L_\Phi(\mathbb{R})$ we have

$$H_\delta(H_\delta u)(t) = -\tfrac{1}{\pi}\sum_{k\in Z} \tfrac{(H_\delta u)(t+(k+1/2)\delta)}{k+1/2} = \tfrac{1}{\pi}\sum_{k\in Z}\tfrac{1}{k+1/2}\cdot\tfrac{1}{\pi}\sum_{m\in Z}\tfrac{u(t+(k+m+1)\delta)}{m+1/2}$$

$$= \tfrac{1}{\pi^2}\sum_{k\in Z}\sum_{m\in Z}\tfrac{u(t+(k+m+1)\delta)}{(k+1/2)(m+1/2)} = \tfrac{1}{\pi^2}\sum_{k\in Z}\sum_{n\in Z}\tfrac{u(t+n\delta)}{(k+1/2)(n-k-1/2)}$$

$$= \tfrac{1}{\pi^2}\sum_{n\in Z}\left(\sum_{k\in Z}\tfrac{1}{(k+1/2)(n-k-1/2)}\right)u(t+n\delta). \tag{5}$$

Since for $n = 0$

$$\sum_{k\in Z}\frac{1}{(k+1/2)(n-k-1/2)} = -4\sum_{k\in Z}\frac{1}{(2k+1)^2} = -\pi^2,$$

and for $n \neq 0$

$$\sum_{k\in Z}\frac{1}{(k+1/2)(n-k-1/2)} = \sum_{k\in Z}\frac{1}{n}\left[\frac{1}{k+1/2}+\tfrac{1}{n-k-1/2}\right]$$

$$= \frac{1}{n}\lim_{N\to\infty}\sum_{|k|\le N}\left[\frac{1}{k+1/2}+\tfrac{1}{n-k-1/2}\right] = 0,$$

then equality (4) follows from (5).

4. Approximation of the singular integral with Hilbert kernel in Orlicz spaces

Let $\Phi \in \Delta_2 \cap \nabla_2$. Denote by $L_\Phi(T)$, the space of all measurable, 2π-periodic functions with finite norm $\|\varphi\|_{L_\Phi(T)} = \|\varphi\|_{L_\Phi([-\pi,\pi])}$.

It is well known that the singular integral with Hilbert kernel

$$(S\varphi)(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\cot\frac{t-\tau}{2}\varphi(\tau)d\tau,$$

is a bounded map in the space $L_\Phi(T)$ (see [45]).

Consider in $L_\Phi(T)$ the sequence of operators

$$(S_n\varphi)(t) = \frac{1}{n}\sum_{k=0}^{n-1}\cot\left(-\tfrac{\pi(2k+1)}{2n}\right)\varphi\left(t+\tfrac{\pi(2k+1)}{n}\right),\quad n\in N.$$

It is easy to verify that if

$$\varphi(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} (a_m \cos mt + b_m \sin mt),$$

then

$$(S_n\varphi)(t) = \sum_{m=1}^{\infty} \lambda_m^{(n)} (a_m \cos mt + b_m \sin mt),$$

where $\lambda_m^{(n)} = 1$ for $m = \overline{1, n-1}$, $\lambda_n^{(n)} = \lambda_{2n}^{(n)} = 0$, $\lambda_m^{(n)} = -1$ for $m = \overline{n+1, 2n-1}$ and $\lambda_{m+2n}^{(n)} = \lambda_m^{(n)}$ for $m \in Z$. It follows from here that for any trigonometric polynomial $P(t)$ of order at most $n-1$

$$(S_n P)(t) = (SP)(t).$$

In [3], the authors prove that the operators S_n are uniformly bounded in $L_p(T)$, $1 < p < \infty$, and for any $n \in N$ the inequality

$$\|S_n\|_{L_p(T) \to L_p(T)} \leq 4 + 2\|\tilde{h}\|_{l_p \to lp}$$

holds. Then it follows from Marcinkiewicz interpolation theorem in Orlicz spaces (see: [22]) that if $\Phi \in \Delta_2 \cap \nabla_2$, then the operators S_n are also uniformly bounded maps in the space $L_\Phi(T)$, that is there exist $\widetilde{C_\Phi} > 0$ such that for any $n \in \mathbb{N}$ and for any $\varphi \in L_\Phi(T)$

$$\|S_n\varphi\|_{L_\Phi(T)} \leq \widetilde{C_\Phi} \|\varphi\|_{L_\Phi(T)}.$$

In [6], it is proved that if $\Phi \in \Delta_2 \cap \nabla_2$, then the sequence of operators S_n strongly converges to the operator S in $L_\Phi(T)$ and for any $\varphi \in L_\Phi(T)$ the following estimate holds:

$$\|S\varphi - S_n\varphi\|_{L_\phi(T)} \leq \left(\|S\|_{L_\Phi(T) \to L_\Phi(T)} + \widetilde{C_\Phi} \right) \cdot E_{n-1}^{\Phi}(\varphi), \ n \in \mathbb{N}, \tag{6}$$

where $E_{n-1}^{\Phi}(\varphi)$ – is the best approximation of the function φ in the metric $L_\Phi(T)$ by trigonometric polynomials of order at most $n-1$.

5. Approximation of the Hilbert transform in Orlicz spaces

Consider the regular integral operator

$$(\mathrm{K}\varphi)(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} K(t,\tau)\varphi(\tau)d\tau, \quad t \in [-\pi, \pi],$$

and the sequence of operators

$$(\mathrm{K}_n\varphi)(t) = \frac{1}{n} \sum_{k=0}^{n-1} K\left(t, t + \tfrac{\pi(2k+1)}{n}\right) \varphi\left(t + \tfrac{\pi(2k+1)}{n}\right), \quad t \in [-\pi, \pi], \quad n \in \mathbb{N},$$

where $K(t,\tau)$ is a continuous function on $[-\pi, \pi]^2$ and $K(t,\tau) = K(t, \tau - 2\pi)$ for $(t,\tau) \in [-\pi, \pi] \times (\pi, 3\pi)$.

Lemma 1. *Let $\Phi \in \Delta_2 \cap \nabla_2$. Then the sequence of operators $\{\mathrm{K}_n\}$ strongly converges to the operator* K *in $L_\Phi(T)$.*

Proof. First assume that $K(t,\tau)$ is a 2π - periodic function by τ . Denote

$$\|K\|_\infty = \max_{t,\tau\in[-\pi,\pi]} |K(t,\tau)|, \quad E_n(K) = \inf \|K - \Phi_n\|_\infty,$$

where $\Phi_n(t,\tau) = \frac{\alpha_0(t)}{2} + \sum_{m=1}^{n}(\alpha_m(t)\cos m\tau + \beta_m(t)\sin m\tau)$, and infimum is taken over all trigonometric polynomials $\alpha_m(t)$, $m = \overline{0,n}$, $\beta_m(t)$, $m = \overline{1,n}$ of order at most n.

Denote $n_0 = \left[\frac{n-1}{2}\right]$. Suppose that

$$q_{n_0}(t) = \frac{a_0}{2} + \sum_{m=1}^{n_0}(a_m \cos mt + b_m \sin mt)$$

and

$$\Phi^{(0)}_{n_0}(t,\tau) = \frac{\alpha^{(0)}_0(t)}{2} + \sum_{m=1}^{n_0}(\alpha^{(0)}_m(t)\cos m\tau + \beta^{(0)}_m(t)\sin m\tau)$$

are the best approximations of the functions φ and K by trigonometric polynomials of order at most n_0, that is

$$\|\varphi - q_{n_0}\|_{L_\Phi} = \inf \|\varphi - P_{n_0}\|_{L_\Phi},$$

where infimum is taken over all trigonometric polynomials P_{n_0} of order at most n_0 and

$$\|K - \Phi^{(0)}_{n_0}\|_\infty = E_n(K).$$

For any trigonometric polynomial $r_{n-1}(t)$ of order at most $n-1$, the equality

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} r_{n-1}(\tau)d\tau = \frac{1}{n}\sum_{k=0}^{n-1} r_{n-1}\left(t + \tfrac{\pi(2k+1)}{n}\right)$$

holds. Therefore

$$(\mathrm{K}\varphi)(t) - (\mathrm{K}_n\varphi)(t) = (\mathrm{K}-\mathrm{K}_n)(\varphi - q_{n_0})(t) + \frac{1}{2\pi}\int_{-\pi}^{\pi}\left[K(t,\tau) - \Phi^{(0)}_{n_0}(t,\tau)\right] q_{n_0}(\tau)d\tau$$

$$+\frac{1}{n}\sum_{k=0}^{n-1}\left[K(t,t+\tau^{(n)}_k) - \Phi^{(0)}_{n_0}(t,t+\tau^{(n)}_k)\right] q_{n_o}(t+\tau^{(n)}_k),$$

where $\tau^{(n)}_k = \frac{\pi(2k+1)}{n}$, $k \in Z$. It follows from here and from inequalities

$$\|\mathrm{K}\|_{L_\Phi(T)\to L_\Phi(T)} \le \|K\|_\infty, \quad \|\mathrm{K}_n\|_{L_\Phi(T)\to L_\Phi(T)} \le \|K\|_\infty$$

that

$$\|\mathrm{K}\varphi - \mathrm{K}_n\varphi\|_{L_\Phi(T)} \le 2\|K\|_\infty E^{\Phi}_{n_0}(\varphi) + 2E_{n_0}(K)\left[\|\varphi\|_{L_\Phi(T)} + E^{\Phi}_{n_0}(\varphi)\right].$$

This completes the proof of the lemma in this case. Now consider the general case.

Let $\varphi \in L_\Phi(T)$ and $\varepsilon > 0$. Since $\varphi \in L_\Phi(T) \subset L_1(T)$, then it follows from Lebesgue theorem that there exist $\delta_\varepsilon^{(0)} > 0$ such that for any $0 < \delta < \delta_\varepsilon^{(0)}$

$$\int_{\pi-\delta}^{\pi} |\varphi(\tau)| d\tau < \frac{\pi\varepsilon}{4(\|K\|_\infty + 1)} \cdot \Phi^{-1}\left(\frac{1}{2\pi}\right).$$

Denote

$$K^*(t,\tau) = K(t,\tau) \quad \text{for} \quad (t,\tau) \in [-\pi,\pi] \times [-\pi, \pi - \delta_\varepsilon],$$

$$K^*(t,\tau) = K(t, \pi - \delta_\varepsilon) + \frac{\tau - \pi + \delta_\varepsilon}{\delta_\varepsilon}[K(t,-\pi) - K(t, \pi - \delta_\varepsilon)]$$
$$\text{for} \quad (t,\tau) \in [-\pi,\pi] \times [\pi - \delta_\varepsilon, \pi],$$

$$K^*(t, \tau + 2\pi) = K^*(t,\tau) \quad \text{for any } (t,\tau) \in [-\pi,\pi] \times \mathbb{R},$$

where $\delta_\varepsilon = \min\left\{\delta_\varepsilon^{(0)}, \frac{\pi\varepsilon}{8\|K\|_\infty \|\varphi\|_{L_\Phi(T)}}, 1\right\}$.

Since the function $K^*(t,\tau)$ is continuous and 2π-periodic by τ, the sequence of operators

$$(\mathrm{K}_n^*\varphi)(t) = \frac{1}{n}\sum_{k=0}^{n-1} K^*(t, t + \tau_k^{(n)})\varphi(t + \tau_k^{(n)}), \quad t \in [-\pi,\pi], \quad n \in \mathbb{N}$$

strongly converges to the operator

$$(\mathrm{K}^*\varphi)(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} K^*(t,\tau)\varphi(\tau)d\tau$$

in $L_\Phi(T)$.Therefore, the inequality

$$\|\mathrm{K}_n^*\varphi - \mathrm{K}^*\varphi\|_{L_\Phi(T)} < \varepsilon/2$$

is satisfied for large values of n. Moreover, since for any $t \in [-\pi,\pi]$

$$|(\mathrm{K}\varphi)(t) - (\mathrm{K}^*\varphi)(t)| \leq \frac{1}{2\pi}\int_{\pi-\delta_\varepsilon}^{\pi} |K(t,\tau) - K^*(t,\tau)||\varphi(\tau)|d\tau$$
$$\leq \frac{\|K\|_\infty}{\pi}\int_{\pi-\delta_\varepsilon}^{\pi} |\varphi(\tau)|d\tau < \frac{\varepsilon}{4} \cdot \Phi^{-1}\left(\frac{1}{2\pi}\right),$$

then

$$\|\mathrm{K}\varphi - \mathrm{K}^*\varphi\|_{L_\Phi(T)} \leq \frac{\varepsilon}{4}.$$

For $n \geq \frac{16\|K\|_\infty \|\varphi\|_{L_p(T)}}{\varepsilon}$ we have

$$\|\mathrm{K}_n\varphi - \mathrm{K}_n^*\varphi\|_{L_\Phi(T)} \leq \frac{1}{n} \cdot \left(\frac{n}{2\pi} \cdot \delta_\varepsilon + 1\right) \cdot 2\|K\|_\infty \|\varphi\|_{L_\Phi(T)} \leq \frac{\varepsilon}{4}.$$

Therefore for sufficiently large values n

$$\|\mathrm{K}_n\varphi - \mathrm{K}\varphi\|_{L_\Phi(T)}$$
$$\leq \|\mathrm{K}_n\varphi - \mathrm{K}_n^*\varphi\|_{L_\Phi(T)} + \|\mathrm{K}_n^*\varphi - \mathrm{K}^*\varphi\|_{L_\Phi(T)} + \|\mathrm{K}^*\varphi - \mathrm{K}\varphi\|_{L_\Phi(T)} < \varepsilon.$$

Corollary 1. *The sequence of operators*

$$(\widetilde{\mathrm{K}}_n\varphi)(t) = \frac{1}{n}\sum_{\{k\in\mathbb{Z}: t+\tau_k^{(n)}\in[-\pi,\pi]\}} K(t, t+\tau_k^{(n)})\varphi(t+\tau_k^{(n)}),\ t\in[-\pi,\pi],\ n\in\mathbb{N}$$

strongly converges to the operator K *in* $L_\Phi([-\pi,\pi])$.

Corollary 2. *If the function $K(t,\tau)$ is continuous on $[\pi m, \pi m + 2\pi q]\times[-\pi,\pi]$, then for any $\varphi\in L_\Phi(T)$ the sequence of functions*

$$(\widetilde{\mathrm{K}}_n\varphi)(t) = \frac{1}{n}\sum_{\{k\in\mathbb{Z}: t+\tau_k^{(n)}\in[-\pi,\pi]\}} K(t, t+\tau_k^{(n)})\varphi(t+\tau_k^{(n)}),\ t\in[\pi m, \pi m+2\pi q],$$

converges to the function

$$(\mathrm{K}\varphi)(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} K(t,\tau)\varphi(\tau)d\tau,\quad t\in[\pi m, \pi m+2\pi q]$$

in $L_\Phi([\pi m, \pi m+2\pi q])$, where $m\in\mathbb{Z}$, $q\in\mathbb{N}$.

Corollary 3. *If the function $K_0(t)$ is continuous on $[-\pi,\pi]$, then the sequence of operators*

$$(\mathrm{K}_n^0\varphi)(t) = \frac{1}{2n}\sum_{k=-n}^{n-1} K_0\left(\tfrac{\pi(2k+1)}{2n}\right)\varphi\left(t+\tfrac{\pi(2k+1)}{2n}\right),\ t\in[-\pi,\pi],\ n\in\mathbb{N}$$

strongly converges to the operator

$$(\mathrm{K}^0\varphi)(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} K_0(\tau)\varphi(t+\tau)d\tau,\quad t\in[-\pi,\pi]$$

in $L_\Phi(T)$.

In the following theorem we prove that for any $\delta>0$ the sequence of operators $\{H_{\delta/n}\}_{n\in\mathbb{N}}$ strongly converges to the operator H in $L_\Phi(\mathbb{R})$.

Theorem 3. *Let $\Phi\in\Delta_2\cap\nabla_2$. Then for any $\delta>0$ the sequence of the operators $\{H_{\delta/n}\}_{n\in\mathbb{N}}$ strongly converges to the operator H in $L_\Phi(\mathbb{R})$, that is for any $u\in L_\Phi(\mathbb{R})$ the following inequality holds:*

$$\lim_{n\to\infty}\|H_{\delta/n}u - Hu\|_{L_\Phi(\mathbb{R})} = 0.$$

Proof. We have divided the proof into three steps.

Step 1. Let us first prove that the operator

$$(H^*\varphi)(t) = \frac{1}{\pi}\int_{t-\pi}^{t+\pi}\frac{\varphi(\tau)}{t-\tau}d\tau$$

is a bounded operator in $L_\Phi(T)$. Indeed, for any $\varphi \in L_\Phi(T)$ we have

$$(H^*\varphi)(t) = \frac{1}{\pi}\int_{t-\pi}^{t+\pi} \tfrac{\varphi(\tau)}{t-\tau} d\tau = \frac{1}{\pi}\int_{t-\pi}^{t+\pi} \left[\tfrac{1}{t-\tau} - \tfrac{1}{2}\cot\tfrac{t-\tau}{2}\right] \varphi(\tau) d\tau + (S\varphi)(t)$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} \left[\cot\tfrac{\tau}{2} - \tfrac{2}{\tau}\right] \varphi(t+\tau) d\tau + (S\varphi)(t). \tag{7}$$

Since the function

$$K_0(\tau) = \cot\frac{\tau}{2} - \frac{2}{\tau} \quad \text{for} \quad \tau \neq 0, \quad K_0 = 0$$

is continuous on $[-\pi,\pi]$, then it follows from (7) that the operator H^* is bounded in $L_\Phi(T)$.

Consider the sequence of operators

$$(H^*_n\varphi)(t) = \frac{1}{\pi}\sum_{k=-n}^{n-1} \tfrac{1}{-k-1/2}\varphi\left(t + \tfrac{\pi(2k+1)}{2n}\right),\ t \in [-\pi,\pi],\ n \in \mathbb{N}.$$

Since for any $\varphi \in L_\Phi(T)$

$$(H^*_n\varphi)(t) = \frac{1}{2n}\sum_{k=-n}^{n-1} \left[\cot\left(\tfrac{\pi(2k+1)}{4n}\right) - \tfrac{4n}{\pi(2k+1)}\right]\varphi\left(t + \tfrac{\pi(2k+1)}{2n}\right) + (S_{2n}\varphi)(t)$$

$$= \frac{1}{2n}\sum_{k=-n}^{n-1} K_0\left(\tfrac{\pi(2k+1)}{2n}\right)\varphi\left(t + \tfrac{\pi(2k+1)}{2n}\right) + (S_{2n}\varphi)(t),$$

then it follows from (6) and from Corollary 3 that the sequence of operators H^*_n strongly converges to the operator H^* in $L_\Phi(T)$.

Step 2. Let us prove that the sequence of operators

$$(H_{\pi/(4n)}u)(t) = \frac{1}{\pi}\sum_{k\in Z} \tfrac{1}{-k-1/2} u\left(t + \tfrac{\pi(k+1/2)}{4n}\right),\ t \in \mathbb{R},\ n \in \mathbb{N}$$

strongly converges to the operator H in $L_\Phi(\mathbb{R})$. At first assume that *supp* $u \subset [-\pi/4,\pi/4]$. Denote by φ 2π-periodic function, coinciding with the function u on $[-\pi/4,\pi/4]$ and equal to zero in $[-\pi,\pi] \setminus [-\pi/4,\pi/4]$. Since for any $t \in [-\pi/2,\pi/2]$

$$(Hu)(t) = \frac{1}{\pi}\int_{-\pi/4}^{\pi/4} \tfrac{u(\tau)}{t-\tau} d\tau = (H^*\varphi)(t), \tag{8}$$

$$(H_{\pi/n}u)(t) = \frac{1}{\pi}\sum_{k=-n}^{n-1} \tfrac{1}{-k-1/2} u\left(t + \tfrac{\pi(k+1/2)}{n}\right)$$

$$= \frac{1}{\pi}\sum_{k=-n}^{n-1} \tfrac{1}{-k-1/2}\varphi\left(t + \tfrac{\pi(k+1/2)}{n}\right) = (H^*_n\varphi)(t), \tag{9}$$

and the sequence of operators H_n^* strongly converges to the operator H^* in $L_\Phi(T)$, then it follows from (8) and (9) that for any $\varepsilon > 0$ for large values of n

$$\|H_{\pi/n}u - Hu\|_{L_\Phi([-\pi/2,\pi/2])} = \|H_n^*\varphi - H^*\varphi\|_{L_\Phi([-\pi/2,\pi/2])}$$
$$\leq \|H_n^*\varphi - H^*\varphi\|_{L_\Phi(T)} < \varepsilon. \tag{10}$$

Due to the inequalities

$$|(Hu)(t)| \leq \frac{1}{\pi}\int_{-\pi/4}^{\pi/4}\left|\frac{u(\tau)}{t-\tau}\right| d\tau \leq \frac{\|u\|_{L_1([-\pi/4,\pi/4])}}{\pi(|t|-\pi/4)},\ |t| > \pi/4,$$

$$|(H_{\pi/n}u)(t)| \leq \frac{1}{\pi}\sum_{k\in\mathbb{Z}_{(n)}^{(t)}} \frac{1}{|k+1/2|}\left|u\left(t+\tfrac{\pi(k+1/2)}{n}\right)\right|$$
$$\leq \frac{1}{n(|t|-\pi/4)}\sum_{k\in\mathbb{Z}_{(n)}^{(t)}}\left|u\left(t+\tfrac{\pi(k+1/2)}{n}\right)\right|, \quad |t| > \pi/4,$$

where $\mathbb{Z}_{(n)}^{(t)} = \{k \in \mathbb{Z} : t + \frac{\pi(k+1/2)}{n} \in [-\pi/4, \pi/4]\}$, we get that for any $M > 2\pi$

$$\|Hu\|_{L_\Phi([M,\infty))} \leq \frac{\|u\|_{L_1([-\pi/4,\pi/4])}}{\pi}\cdot\|f_0\|_{L_\Phi([M,\infty))},$$

where $f_0(t) = \frac{1}{t-\pi/4} \in L_\Phi([M,\infty))$,

$$\|H_{\pi/n}u\|_{L_\Phi([M,\infty))} \leq \frac{1}{n(M-\pi/4)}\left\|\sum_{k\in\mathbb{Z}_{(n)}^{(t)}}\left|u\left(\cdot+\tfrac{\pi(k+1/2)}{n}\right)\right|\right\|_{L_\Phi([M,\infty))}$$
$$\leq \frac{1}{n(M-\pi/4)}\cdot n\cdot\|u\|_{L_\Phi([-\pi/4,\pi/4])} = \frac{\|u\|_{L_\Phi([-\pi/4,\pi/4])}}{M-\pi/4}$$

Similar inequalities holds for $\|Hu\|_{L_\Phi((-\infty,-M])}$ and for $\|H_{\pi/n}u\|_{L_\Phi((-\infty,-M])}$. Therefore, for any $\varepsilon > 0$ there exist $m_0 \geq 4$ such that

$$\|Hu\|_{L_\Phi(R\setminus[-\frac{\pi m_0}{2},\frac{\pi m_0}{2}])} < \varepsilon, \quad \|H_{\pi/n}u\|_{L_\Phi(R\setminus[-\frac{\pi m_0}{2},\frac{\pi m_0}{2}])} < \varepsilon. \tag{11}$$

Since the function $\frac{1}{t-\tau}$ is continuous on a rectanhle $[2\pi, 2\pi m_0]\times[-\pi,\pi]$, then it follows from Corollary 2 that the sequence of functions

$$(\mathrm{W}_n\varphi)(t) = \frac{2}{n}\sum_{\{k\in Z: t+\frac{\pi(2k+1)}{n}\in[-\pi,\pi]\}} \frac{\varphi(t+\pi(2k+1)/n)}{-\pi(2k+1)/n}$$
$$= \frac{1}{\pi}\sum_{\{k\in Z: t+\frac{\pi(2k+1)}{n}\in[-\pi,\pi]\}} \frac{\varphi(t+\pi(2k+1)/n)}{-k-1/2}, \quad n \in N$$

converges to the function

$$(\mathrm{W}\varphi)(t) = \int_{-\pi}^{\pi} \frac{\varphi(\tau)}{t-\tau} d\tau$$

in $L_\Phi([2\pi, 2\pi m_0])$. Denote by ψ the function, defined on $[-\pi,\pi]$ by the equality $\psi(\tau) = u(\tau/4)$. Then it follows from the equations

$$(Hu)(t) = \frac{1}{\pi}\int_{-\pi/4}^{\pi/4} \frac{u(\tau)}{t-\tau} d\tau = \frac{1}{\pi}\int_{-\pi}^{\pi} \frac{u(\tau/4)}{4t-\tau} d\tau = (\mathrm{W}\psi)(4t),\ t \in [\pi/2, \pi m_0/2],$$

$$(H_{\pi/(4n)}u)(t) = \frac{1}{\pi} \sum_{k \in Z_{(4n)}^{(t)}} \tfrac{u(t+\pi(k+1/2)/4n)}{-k-1/2}$$

$$= \frac{1}{\pi} \sum_{k \in Z_{(4n)}^{(t)}} \tfrac{\psi(4t+\pi(k+1/2)/n)}{-k-1/2} = (\mathrm{W}_n\psi)(4t),\ t \in [\pi/2, \pi m_0/2],$$

that the sequence of functions $H_{\pi/(4n)}u$ converges to the function Hu in the space $L_\Phi([\pi/2, \pi m_0/2])$. Therefore, for large values of n

$$\|H_{\pi/(4n)}u - Hu\|_{L_\Phi([\pi/2,\pi m_0/2])} < \varepsilon. \tag{12}$$

Similarly, for large values on n

$$\|H_{\pi/(4n)}u - Hu\|_{L_\Phi([-\pi m_0/2,-\pi/2])} < \varepsilon. \tag{13}$$

It follows from (10)-(13) that in the case *supp* $u \subset [-\pi/4, \pi/4]$

$$\lim_{n\to\infty} \|H_{\pi/(4n)}u - Hu\|_{L_\Phi(R)} = 0. \tag{14}$$

Now suppose that *supp* $u \subset [-\pi m/4, \pi m/4]$ for some $m \in N$. Denote by u_0 the function, defined on $[-\pi/4, \pi/4]$ by the equatin $u_0(t) = u(mt)$. Then for any $t \in R$

$$(Hu)(t) = \frac{1}{\pi}\int_{-\pi m/4}^{\pi m/4} \frac{u(\tau)}{t-\tau} d\tau = \frac{1}{\pi}\int_{-\pi/4}^{\pi/4} \frac{u(m\tau)}{t-\tau} m d\tau = (Hu_0)(t/m),$$

$$(H_{\pi/(4n)}u)(t) = \frac{1}{\pi} \sum_{\{k\in Z: t+\frac{\pi(k+1/2)}{4n} \in [-\frac{\pi m}{4}, \frac{\pi m}{4}]\}} \tfrac{u(t+\pi(k+1/2)/4n)}{-k-1/2}$$

$$= \frac{1}{\pi} \sum_{k \in Z_{(4mn)}^{(t/m)}} \tfrac{u_0(t/m+\pi(k+1/2)/(4mn))}{-k-1/2} = (H_{\pi/(4mn)}u_0)(t/m).$$

Since equation (14) holds for u_0, we obtain that

$$\lim_{n\to\infty} \|H_{\pi/(4n)}u - Hu\|_{L_\Phi(R)} = m^{1/p} \lim_{n\to\infty} \|H_{\pi/(4mn)}u_0 - Hu_0\|_{L_\Phi(R)} = 0.$$

Now consider the general case. Let us prove that equation (14) holds for any $u \in L_\Phi(R)$. For any $u \in L_\Phi(R)$ and $\varepsilon > 0$ there exist $m \in N$ such that

$$\|u - u_m\|_{L_\Phi(R)} < \varepsilon, \tag{15}$$

where $u_m(t) = u(t) \cdot \chi_{[-\pi m/4, \pi m/4](t)}$. Since equation (14) holds for u_m, and it follows from (3), (15) that

$$\|H_{\pi/(4n)}(u - u_m) - H(u - u_m)\|_{L_\Phi(R)}$$
$$\leq \left[\|H_{\pi/(4n)}\|_{L_\Phi(R)\to L_\Phi(R)} + \|H\|_{L_\Phi(R)\to L_\Phi(R)}\right] \cdot \|u - u_m\|_{L_\Phi(R)}$$
$$\leq \varepsilon \cdot \left[C_\Phi + \|H\|_{L_\Phi(R)\to L_\Phi(R)}\right],$$

then we get that the equation (14) also holds for the function u.

Step 3. Let us prove that for any $\delta > 0$ the secuence of the operators $\{H_{\delta/n}\}_{n\in N}$ strongly converges to the operator H in $L_\Phi(R)$. Let $u \in L_\Phi(R)$. Denote $w(t) = u(4\delta t/\pi)$, $t \in R$. Then for any $t \in R$

$$(Hu)(t) = \frac{1}{\pi} \int_R \tfrac{u(\tau)}{t-\tau} d\tau = \frac{1}{\pi} \int_R \tfrac{w(\pi\tau/(4\delta))}{t-\tau} d\tau$$
$$= \frac{1}{\pi} \int_R \tfrac{w(\tau)}{\pi t/(4\delta)-\tau} d\tau = (Hw)(\pi t/(4\delta)), \tag{16}$$
$$(H_{\delta/n}u)(t) = \frac{1}{\pi} \sum_{k\in Z} \tfrac{u(t+(k+1/2)\delta/n)}{-k-1/2}$$
$$= \frac{1}{\pi} \sum_{k\in Z} \tfrac{w(\pi t/(4\delta)+\pi(k+1/2)/(4n))}{-k-1/2} = (H_{\pi/(4n)}w)(\pi t/(4\delta)). \tag{17}$$

Since $\lim_{n\to\infty} \|H_{\pi/(4n)}w - Hw\|_{L_\Phi(R)} = 0$, then it follows from (16), (17) that

$$\lim_{n\to\infty} \|H_{\delta/n}u - Hu\|_{L_\Phi(R)} = 0.$$

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Asymptotic Behavior of Eigenvalues of a Boundary Value Problem for Laplace Equation

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Abstract. In the square $\Omega = (0, 2\pi) \times (0, 2\pi)$ we consider a spectral boundary value problem for the Laplace equation in the case when one of the boundary conditions contain mixed derivatives on the line $x = 2\pi$, when $y \in [0, 2\pi]$.
It is shown that the considered spectral problem has two series of eigenvalues one of which is finite, the another one asymptotically behaves as $\lambda_{n,k} \sim k^2 + \frac{n^2}{4} \quad (k, n \to \infty)$.

Key Words and Phrases: eigenvalues, Hilbert space, Laplace equations, asymptotic formula.

2010 Mathematics Subject Classifications: 35G25, 35P20, 47A75

1. Introduction

It is known that in the bounded domain there exist spectral problems for the Laplace equation (for example, first, second, third) in which eigenvalues are nonnegative, discrete and have a unique limit point $+\infty$. It this case, it is customary to say that eigenvalues of spectral boundary value problems for the Laplace equation behave as classic. These exist spectral boundary value problems for the Laplace equation in which the classic nature of eigenvalues is violated. Non-classic nature of eigenvalues of boundary value problems for the Laplace equation appear mainly in two moments:

I. Boundary conditions contain a linear operator (differential or non-differential)

II. One and the same spectral parameter is involved in the equation and in the boundary conditions.

We cite some papers where the classic nature of the eigenvalues of boundary value problems for the Laplace equation is violated.

In [1] V.A. Il'in and A.F. Filippov consider such a spectral problem for the Laplace equation where any point on the positive axis is a condensation point, i.e. classic nature of eigenvalues is violated. In this paper, one of the boundary conditions contains a non-differential operator.

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In S.Ya. Yakubov is work [2], asymptotic behavior of eigenvalues of the following boundary value problems is studied for the Laplace equation in the square $\Omega = (0, 2\pi) \times (0, 2\pi)$, containing in one of the boundary conditions the differential operator.

$$-\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = \lambda u\,(x, y)\,, \tag{1}$$

$$u(0, y) = 0,\ \ \frac{\partial u\,(2\pi, y)}{\partial x} + i\frac{\partial u\,(2\pi, y)}{\partial y} = 0,\ \ y \in [0, 2\pi]\,, \tag{2}$$

$$u(x,\,0) = u(x,\,2\pi),\quad \frac{\partial u\,(x, 0)}{\partial y} = \frac{\partial u\,(x, 2\pi)}{\partial y},\ \ x \in [0, 2\pi] \tag{3}$$

It is proved that for boundary value problems (1)- (3) there exists a sequence of eigenvalues $\{\lambda_k\}$ convergent to zero. More exactly, the classic case on the behavior of eigenvalues for the boundary value problems (1)-(3) is violated .

In A.N. Kozhevnikov's work [3], in the boundary domain. $.\Omega \in R^n$ with a rather smooth boundary Γ for the Laplace equation the following spectral problem is studied:

$$-\Delta u = \lambda u \ \text{ in } \ \Omega, \tag{4}$$

$$-u = \lambda\frac{\partial u}{\partial \nu} \ \text{ in } \ \Gamma, \tag{5}$$

where λ is a spectral parameter, ν is an interior normal to the boundary Γ. It is proved that the spectrum of the boundary value problem (4), (5) is discrete and consists of two series of eigenvalues convergent to zero and to $+\infty$, respectively. More exactly, the eigenvalues of spectral boundary value problem (4), (5) behave as non- classic .

It is known that many boundary value problems for the Laplace equation given in a rectangle are reduced to spectral boundary value problems for second order elliptic differential operator equations in some Hilbert space.

Usually, in spectral problems for elliptic differential operator equations, the operator appearing in the equation is an operator with a discrete spectrum, the eigenevalues of which form a complete orthonormal basis in the same Hilbert space. After spectral expansion in eigenelements of the operator appearing in the equation, the spectral problem stated for elliptic differential operator equations is reduced to a spectral boundary value problem for ordinary differential equations with respect to Fourier coefficients.

In the present paper, using the above method we study asymptotic behavior of eigen values of the following boundary value problems for a two-dimensional Laplace equation in the square $\Omega = (0, 2\pi) \times (0, 2\pi)$:

$$-\frac{\partial^2 u(x, y)}{\partial x^2} - \frac{\partial^2 u(x, y)}{\partial y^2} = \lambda u\,(x, y)\,, \tag{6}$$

$$u(0, y) = 0,\ \ u\,(2\pi, y) + i\frac{\partial^2 u(2\pi, y)}{\partial y \partial x} = 0,\ \ y \in [0, 2\pi]\,, \tag{7}$$

$$u(x,\,0)=\,u\,(x,2\pi)\,,\quad \frac{\partial u(x,\,0)}{\partial y}=\frac{\partial u(x,\;2\pi)}{\partial y},\;\;x\in[0,2\pi]\,. \tag{8}$$

where i is an imaginary number.

It is proved that the eigenvalues of boundary value problem (6)-(8) are real. Then it is shown that problem (6)-(8) has two series of eigenvalues one of which consist of a finitely many numbers, more exactly of six eigenvalues satisfying the inequality : $k^2-1<\lambda_k<k^2$, $k=1,...,6$; the another one behaves as $\lambda_{n,k}\sim k^2+\frac{n^2}{4}$; $k,n\in N$. In other works, the classicsum of eigenvalues of or boundary value problems (6)-(8) is violeted.

By the above [4-7] studies asymptotic behavior of eigenvalues of boundary values problems for second order elliptic differential operator equations in the case one and the same spectral parameter is involved in the equation and in the boundary condition. It is proved that in [4-7] the eigenvalues of the problem under consideration behave with non-classic asymptotics.

2. Asymptotic formulas for eigenvalues

Definition 1. *Am eigenfunctions of the problem* (6)-(8) *is a identically non-zero function* $u(x,y)\in C^{(2)}\left(\overline{\Omega}\right)$ *that on the boundary* Ω *satisfy boundary conditions* (7), (8) *and for some* λ *satisfies on* Ω *the equation* (6). *The values of* λ *for which there exist eigenfunctions, are called eigenvalues of problem* (6)-(8).

Lemma 1. *The eigenvalues of the boundary value of problem* (6)-(8) *are real.*

Proof. In Hilbert space $L_2\,(0,2\pi)$ we consider the operators A and B, determined by the following equalities:

$$D(A):=W_2^2\,\left((0.2\pi)\,,u\,(0)=u\,(2\pi)\,,u'\,(0)=u'\,(2\pi)\right),\;Au=-\frac{d^2u}{dy^2};$$

$$D(B):=W_2^1\,((0,2\pi)\,;\;u(0)=u\,(2\pi))\,,\;\;Bu=i\frac{du}{dy}.$$

The eigenvalues of the operator A are the numbers $\mu_k(A)=k^2$, $k=0,1,...\infty$ that for $k>0$ correspond to the pair of eigenfunctions

$$u_{k1}(y)=\frac{1}{\sqrt{2\pi}}e^{iky},\;u_{k2}(y)=\frac{1}{\sqrt{2\pi}}e^{-iky},\;k=1,2,...\infty.$$

The eigenvalues $\mu_k(B)=-k$, $k=0,1,...,\infty$ correspond to the pair of eigenfunction $u_{k1}(y)$, the eigenvalues $\mu_{-k}(B)=k$, $k=1,2,...,\infty$, correspond to the eigenfunctions $u_{k2}(y)$.

The problem (6)-(8) is equivalent to the problem

$$-u''(x)+Au(x)=\lambda u(x),\;x\in(0,2\pi)\,, \tag{9}$$

$$u(0)=0,\quad u\,(2\pi)+Bu'\,(2\pi)=0, \tag{10}$$

where $u(x)$ is a vector function with the values from $L_2\,(0,2\pi)$. It is known that the system $\{u_{k1}\,(y)\,,\ u_{k2}\,(y)\}$ $k=\overline{1,\infty}$ forms a complete orthonormed basis in $L_2\,(0,2\pi)$. Then almost everywhere on $(0,1)$ any element $u(x)\in L_2\,(0,2\pi)$ expands in Fourier series:

$$u(x)=\sum_{k=1}^{\infty}\left[(u(x),u_{k1})\,u_{k1}+(u(x),u_{k2})\,u_{k2})\right].$$

If $Au(x)\in L_2\,(0,2\pi)\,,\ \ u''(x)\in L_2\,(0,2\pi)$, then almost everywhere on $(0,2\pi)$ we have the expansion

$$Au(x)=\sum_{k=1}^{\infty}k^2\left[(u(x),u_{k1})\,u_{k1}+(u(x),u_{k2})\,u_{k2}\right],$$

$$u''(x)=\sum_{k=1}^{\infty}\left[\big(u''(x),u_{k1}\big)\,u_{k1}+\big(u''(x),u_{k2}\big)\,u_{k2}\right].$$

If $u(x)\in D\,(B)$, then almost everywhere on $(0,2\pi)$ we have the expansion
$u(x)=\sum\limits_{k=1}^{\infty}\left[-k\,(u(x),u_{k1})\,u_{k1}+k\,(u(x),u_{k2})\,u_{k2}\right]$.

Taking into account these spectral expansions in problem (9), (10) for the Fourier coefficients $\tilde{u}_{k1}(x)=(u(x),u_{k1})$ we obtain the problem

$$-\tilde{u}''_{k1}+k^2\tilde{u}_{k1}(x)=\lambda\tilde{u}_{k1}(x),\ \ x\in(0,2\pi)\,,\tag{11}$$

$$\tilde{u}_{k1}\,(0)=0,\ \ \tilde{u}_{k1}\,(2\pi)-k\tilde{u}'_{k1}(2\pi)=0,\tag{12}$$

and for the coefficients $\tilde{u}_{k2}(x)=(u(x),u_{k2})$ we obtain the boundary value problem

$$-\tilde{u}''_{k2}(x)+k^2\tilde{u}_{k2}(x)=\lambda\tilde{u}_{k2}(x),\ \ x\in(0,2\pi)\,,\tag{13}$$

$$\tilde{u}_{k2}\,(0)=0,\ \ \tilde{u}_{k2}\,(2\pi)+k\tilde{u}'_{k2}(2\pi)=0.\tag{14}$$

Thus, the study of eigenvalues of boundary value problems (6)-(8) is reduced to the study of eigenvalues of boundary value problems (11), (12) and (13), (14).

We show that the eigenvalues of problem (11), (12) are real.

Muitiplying the equation (11) by the function $\overline{\tilde{u}}_{k1}\,(x)$ and integrating on the interval $(0,2\pi)\,,$ we obtain the equality

$$-\int_0^{2\pi}\tilde{u}''_{k1}\,(x)\,\overline{\tilde{u}}_{k1}\,(x)\,dx+k^2\int_0^{2\pi}|\tilde{u}_{k1}\,(x)|^2\,dx=\lambda\int_0^{2\pi}|\tilde{u}_{k1}\,(x)|^2\,dx.$$

Integrating by parts, we obtain :

$$-\overline{\tilde{u}}_{k1}\,(2\pi)\,\tilde{u}'_{k1}\,(2\pi)+\overline{\tilde{u}}_{k1}\,(0)\,\tilde{u}'_{k1}\,(0)+\int_0^{2\pi}\left|\tilde{u}'_{k1}\,(x)\right|^2dx+$$

$$+k^2\int_0^{2\pi}|\tilde{u}_{k1}\,(x)|^2\,dx=\lambda\int_0^{2\pi}|\tilde{u}_{k1}\,(x)|^2\,dx.\tag{15}$$

Using boundary conditions (12) from equality (15) we obtain

$$-\frac{1}{k}\left|\tilde{u}_{k1}\left(2\pi\right)\right|^2+\int_0^{2\pi}\left|\tilde{u}_{k1}'\left(x\right)\right|^2dx+k^2\int_0^{2\pi}\left|\tilde{u}_{k1}\left(x\right)\right|^2dx=\lambda\int_0^{2\pi}\left|\tilde{u}_{k1}\left(x\right)\right|^2dx.$$

Hence it follows that the eigenvalues of problem (11), (12) are real. ◄

In a similar way it is shown that the egienvalues of boundary value problem (13), (14) are also real.

Lemma 1 is proved.

Obviously, there must be $\lambda \neq k^2$. Since for $\lambda = k^2$ problem (11), (12) has only a trivial solution .

Theorem 1. *Boundary value problem* (6)-(8) *has two series of eigenvalues one of which is a finite sequence satisfying the inequality* $k^2-1<\lambda_k<k^2,\ k=\overline{1,6}$, *and the another one behaves asymptotically as* $\lambda_{n,k}\sim k^2+\frac{n^2}{4}$ *for* $k,n\to\infty$.

Proof. The general solution of ordinary differential equation (11) is in the form

$$\tilde{u}_{k1}(x)=c_1e^{-x\sqrt{k^2-\lambda}}+c_2e^{-(2\pi-x)\sqrt{k^2-\lambda}}, \tag{16}$$

where c_i $(i=1,2)$ are arbitrary constants .

Having substituted (16) in (12), we obtain a system with respect to c_i $(i=1,2)$, whose determinant is of the form :

$$D_k\left(\lambda\right)=\left(k\sqrt{k^2-\lambda}+1\right)e^{-4\pi\sqrt{k^2-\lambda}}-\left(k\sqrt{k^2-\lambda}-1\right).$$

The eigenvalues of boundary value problem (11), (12) consist of real $\lambda\neq k^2$, that even if for one k satisfy the transcendental equation:

$$\left(k\sqrt{k^2-\lambda}+1\right)e^{-4\pi\sqrt{k^2-\lambda}}-\left(k\sqrt{k^2-\lambda}-1\right)=0. \tag{17}$$

Equation (17) is equivalent to the equation

$$k\sqrt{k^2-\lambda}ch\left(2\pi\sqrt{k^2-\lambda}\right)-sh\left(2\pi\sqrt{k^2-\lambda}\right)=0. \tag{18}$$

Let us write equation (18) in the form

$$k\sqrt{k^2-\lambda}-th\left(2\pi\sqrt{k^2-\lambda}\right)=0. \tag{19}$$

We find the eigenvalues of boundary value problem (11), (12) for which $\lambda<k^2,\ k\in N$. Let us put to equation (19), $2\pi\sqrt{k^2-\lambda}=y$ $(0<y<2\pi k)$. Then equation (19) takes the form

$$thy-\frac{k}{2\pi}y=0,\ 0<y<2\pi k,\ \ k=1,2,...\infty. \tag{20}$$

We note that the roots of equation (20) are the abscissa of intersection point of the graphs of the function $f_k(y) = thy,\ y \in (0, 2\pi k)$ and $g_k(y) = \frac{k}{2\pi}y,\ y \in (0, 2\pi k)\,,\ k = 1, 2, ...\infty$.

Show that for each fixed $k = \overline{1, 6}$ the equation (20) has a unique solution. Indeed, since the angular coefficient of the tangent drawn to the curve $f_k(y) = thy,\ y \in (0, 2\pi k)$, $k = \overline{1.\infty}$ at the point $y = 0$ equals 1, and the angular coefficient of the straightline $g_k(y) = \frac{k}{2\pi}y,\ y \in (0, 2\pi k)\,$, is less than 1, only for $k = \overline{1,\,6}$. Consequently, the graphs of these functions may intersect for $k = \overline{1,\,6}$. For $k = 7, 8, ..., \infty$ the angular coefficient of the straightline $g_k(y) = \frac{k}{2\pi}y,\ y \in (0, 2\pi k)$ is greater than a unit, therefore we confirm that the graph of the functions $f_k(y) = thy,\ \ y \in (0, 2\pi k)$ and $g_k(y) = \frac{k}{2\pi}y,\ y \in (0, 2\pi k)$ can not intersect for $k = 7, 8, ..., \infty$. Obviously, the intersection points of these functions are located in the open rectangle $(0, 2\pi) \times (0, 1)$.

Consequently, the abscissa of the intersection points of the curve $f_k(y) = thy,\ y \in (0, 2\pi k)$, and of the straightline $g_k(y) = \frac{k}{2\pi}y,\ y \in (0, 2\pi k)$ for $k = \overline{1,\,6}$ are located in the interval $(0, 2\pi)$.

Let us consider the functions

$$\varphi_k(y) = thy - \frac{k}{2\pi}y, 0 < y < 2\pi\,, k = \overline{1,\,6}. \tag{21}$$

For a rather small $\varepsilon > 0$ we have

$$\varphi_k(\varepsilon) = th\varepsilon - \frac{k}{2\pi}\varepsilon > 0,$$

since for small $\varepsilon > 0$, $th\varepsilon \sim \varepsilon$.

$$\varphi_k(2\pi) = \lim_{y \to 2\pi - 0}\left(thy - \frac{k}{2\pi}y\right) = th2\pi - k < 0,$$

since for any $y > 0$, $0 < thy < 1$. In what follows, by virtue of the Cauchy theorem on the zeros of a continuous function, for each $k = \overline{1,\,6}$ the function $\varphi_k(y)$, determined by formula (21) has a unique zero belonging to the interval $(0, 2\pi)$. The uniquesness of zeros of the function $\varphi_k(y)$ for each $k = \overline{1,\,6}$ follows from the fact that the function $f_k(y)$, $y \in (0, 2\pi)$ and the staightline $g_k(y) = \frac{k}{2\pi}y,\ y \in (0, 2\pi)$ are monotonically increasing. We denote the zeros of the function $\varphi_k(y)$ by y_k: $0 < y_k < 2\pi,\ k = \overline{1,\,6}$. Consequently, for each $k = \overline{1,\,6}$, $0 < 2\pi\sqrt{k^2 - \lambda} < 2\pi$. Hence, for each $k = \overline{1,\,6}$ we have

$$\lambda_k > k^2 - 1.$$

On the other hand, we consider such eigenvalues λ, for which $\lambda_k < k^2$. As a result, for the eigenvalues of boundary value problem (11), (12) satisfying the inequality $\lambda < k^2$, we have

$$k^2 - 1 < \lambda_k < k^2, k = \overline{1,\,6}.$$

We now find the eigenvalues of boundary value problems (11), (12) for which $\lambda > k^2$. We put in equation (18) $2\pi\sqrt{\lambda - k^2} = z$, $0 < z < +\infty$. Then equation (18) takes the form

$$\frac{k}{2\pi}z\cos z - \sin z = 0, z \in (0, +\infty)\,. \tag{22}$$

Let $z \neq n\pi, \ n \in N$. In this case equation (22) is equivalent to the equation

$$zctgz - \frac{2\pi}{k} = 0, \ z \in (0, +\infty), \ z \neq n\pi, \ n \in N. \tag{23}$$

Let us consider the function

$$\psi_k(z) = zctgz - \frac{2\pi}{k}, \ z \in (0, +\infty), \ z \neq n\pi, \ \ n \in N.$$

Since at each interval $(n\pi, (n+1)\pi), \ \ n \in N$ the function $\psi_k(z)$ accepts the values from $-\infty$ to $+\infty$, and its derivative

$$\psi_k'(z) = \frac{\sin 2z - 2z}{2\sin^2 z} < 0,$$

then n the given interval for each k,the function $\psi_k(z)$ has only one zero z_{nk}: $n\pi < z_{n,k} < (n+1)\pi, \ n \in N$. For each $k \in N$ we find asymptotic formulas for $z_{n,k}$, as $n \to \infty$. From equation (23) we have

$$ctgz = \frac{2\pi}{kz}, \ \ z \in (0, +\infty), \ z \neq n\pi, \ n \in N. \tag{24}$$

Obviously, the points $z_{n,k}$ are the abscissa of the interaction of the graphs of the function $q_k(z) = \frac{2\pi}{kz}$, $z \in (0, +\infty)$, $k \in N$ and branches of the function $ctgz$. More exactly , $z_{n,k}$ are approximate solutions of equation (23) . From the location of graphs of these functions it is clear that for each k, with increasing n, the points $z_{n,k}$ will approach the points $n\pi$, i.e., $z_{n,k} \sim n\pi$.

Hence and from the equality $2\pi\sqrt{\lambda - k^2} = z$ for the eigenvalues of boundary value problem (11), (12) satisfying the condition $\lambda > k^2$, we have the formula: $\lambda_{n,k} \sim k^2 + \frac{n^2}{4}$.

We now study boundary value problem (13), (14). In the same way we show that the eigenvalues of boundary value problems (13), (14) are also real, and $\lambda = k^2$, $k \in N$ is not an eigenvalue.

The general solution of ordinary differential equation (13) is of the form

$$\tilde{u}_{k2}(x) = c_1 e^{-x\sqrt{k^2-\lambda}} + c_2 e^{-(2\pi - x)\sqrt{k^2-\lambda}}, \tag{25}$$

where $c_i \, (i = 1, 2)$ are arbitrary constants .

Having substituted (25) in (14), we obtain a system with respect to $c_i \, (i = 1, 2)$, whose determinant is of the form

$$F_k(\lambda) = \left(1 - k\sqrt{k^2 - \lambda}\right) e^{-4\pi\sqrt{k^2-\lambda}} - \left(1 + k\sqrt{k^2 - \lambda}\right).$$

The eigenvalues of boundary value problem (13), (14) consist of real $\lambda \neq k^2$, $k \in N$, that for even if one k satisfy the equation

$$F_k(\lambda) = 0. \tag{26}$$

Equation (26) is equivalent to the equation

$$k\sqrt{k^2-\lambda}\,ch\left(2\pi\sqrt{k^2-\lambda}\right)+sh\left(2\pi\sqrt{k^2-\lambda}\right)=0, \tag{27}$$

that for $\lambda < k^2$, $k \in N$ has no solutions. Consequently, boundary value problem (13), (14) has no eigenvalues satisfying the condition $\lambda < k^2$, $k \in N$.

We now look for the solutions of equation (27) , that are greater than k^2. Assume $2\pi\sqrt{\lambda-k^2}=z,\ 0<z<+\infty$. Then we can write equation (27) in the form

$$-\frac{k}{2\pi}z\sin z+\cos z=0,\ \ z\in(0,+\infty)\,. \tag{28}$$

Let $z \neq \frac{\pi}{2}+n\pi,\quad n=0,1,...\infty$. Then equation (28) is equivalent to the equation

$$tgz-\frac{2\pi}{kz}=0,\quad z\in(0,+\infty)\,,z\neq\frac{\pi}{2}+n\pi,\quad n=0,1,...\infty. \tag{29}$$

Let us consider the function

$$\eta_k\left(z\right)=tgz-\frac{2\pi}{kz},\quad z\in(0,+\infty)\,,z\neq\frac{\pi}{2}+n\pi,\quad n=0,1,...\infty.$$

Since at each interval $\left(\pi\left(\frac{1}{2}+n\right),\ \pi\left(\frac{3}{2}+n\right)\right),\quad n=0,1,...\infty$ the function $\eta_k\left(z\right)$ accepts the values from $-\infty$ to $+\infty$, and its derivative

$$\eta_k'\left(z\right)=\frac{1}{\cos^2 z}+\frac{2\pi}{k}>0,\quad z\in(0,+\infty)\,,$$

then in it $\eta_k\left(z\right)$ for each k has only one zero z_{nk}: $\pi\left(\frac{1}{2}+n\right)<z_{n,k}<\pi\left(\frac{3}{2}+n\right)$, $n=0,1,2,...\infty$. For each $k\in N$ we find asymptotic formulas for z_{nk}, as $n\to+\infty$.

Obviously, the points $z_{n,k}$ are the abscissa of the intersection of the function $q_k\left(z\right)=\frac{2\pi}{kz},\quad z\in(0,+\infty)\,,k\in N$ and branches of the function tgz, i.e. $z_{n,k}$ is approximate solution of the equation (29). From the location of the graps of these functions it is clear that for each k, with increasing n, the point z_{nk} will approach the point $n\pi$, i.e., $z_{nk}\sim n\pi$. Hence and from the equality $2\pi\sqrt{\lambda-k^2}=z$ for the eigenvalues of the boundary value problem (13), (14) satisfying the condition $\lambda>k^2$ we have the asymptotic formula $\lambda_{n,k}\sim k^2+\frac{n^2}{4}$. ◀

The Theorem 1 is proved .

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Boundedness of the Discrete Riesz Transform on Discrete Morrey spaces

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Abstract. It is well known that the Riesz transform plays an important role in the theory of harmonic functions. This transform has been well studied on classical Lebesgue, Morrey, Sobolev, Besov, Campanato, etc. spaces. But its discrete version has not been well studied. In this paper, we prove that the discrete Riesz transform is a bounded operator in discrete Morrey spaces.

Key Words and Phrases: Riesz transform, discrete Riesz transform, Morrey spaces, discrete Morrey spaces, boundedness.
2010 Mathematics Subject Classifications: 44A15, 46B45, 42B35

1. Introduction

The j-th Riesz transform of the function $f \in L_p(\mathbb{R}^d)$, $1 \le p < +\infty$ is defined as the following singular integral:

$$R_j(f)(x) = \gamma_{(d)} \cdot \text{v.p.} \int_{\mathbb{R}^d} \frac{x_j - y_j}{|x-y|^{d+1}} f(y)dy$$
$$= \gamma_{(d)} \cdot \lim_{\varepsilon \to 0} \int_{\{y \in \mathbb{R}^d:\ |x-y|>\varepsilon\}} \frac{x_j - y_j}{|x-y|^{d+1}} f(y)dy,$$

where $\gamma_{(d)} = \frac{\Gamma((d+1)/2)}{\pi^{(d+1)/2}}$, $\Gamma(z) = \int_0^{+\infty} t^{z-1} \mathrm{e}^{-t} dt$ is Euler's Gamma function.

It is well known (see [26]) that the Riesz transform plays an important role in the theory of harmonic functions. The boundary values of harmonically conjugate in the upper half space functions are interconnected by the Riesz transform. From the theory of singular integrals (see [6, 26]) it is known that the Riesz transform is a bounded operator in the space $L_p(\mathbb{R}^d)$, $1 < p < \infty$, that is, if $f \in L_p(\mathbb{R}^d)$, then $R_j(f) \in L_p(\mathbb{R}^d)$ and the inequality

$$\|R_j f\|_{L_p} \le C_p \|f\|_{L_p}$$

holds; in the case $f \in L_1(\mathbb{R}^d)$ only the weak inequality holds:

$$m\{x \in R^d:\ |(R_j f)(x)| > \lambda\} \le \frac{C_1}{\lambda} \|f\|_{L_1},$$

where m stands for the Lebesgue measure, C_p, C_1 are constants independent of f.

Denote by $l_p := l_p(Z^d)$, $p \geq 1$, the class of sequences $h = \{h_n\}_{n \in Z^d}$ satisfying the condition

$$\|h\|_{l_p} := \left(\sum_{n \in Z^d} |h_n|^p \right)^{1/p} < \infty,$$

where $Z^d := \{(n_1, ..., n_d) : \ n_k \in Z, \, k = 1, ..., d\}$ and Z is the set of integers.

Let $h = \{h_n\}_{n \in Z^d} \in l_p$, $p \geq 1$. Namely, the sequence $\widetilde{R}_j(h) = \{(\widetilde{R}_j h)_n\}_{n \in Z^d}$ is called the Riesz transform of the sequence h, where

$$(\widetilde{R}_j h)_n = \sum_{m \in Z^d, \, m \neq n} \frac{n_j - m_j}{|n - m|^{d+1}} \cdot h_m, \ n \in Z^d.$$

Note that if $h \in l_p$, $1 \leq p < \infty$, then from the Hölder inequality it follows that the series $\sum_{m \in Z^d, \, m \neq n} \frac{n_j - m_j}{|n-m|^{d+1}} \cdot h_m$ absolutely converges, and therefore the Riesz transform of the sequence h exists.

In [4, 5, 8, 9, 10, 17, 18, 20, 22] the boundedness of the Riesz transform in other function spaces was studied. But discrete version of the Riesz transform has not been well studied. In this paper, we prove that the discrete Riesz transform is a bounded operator on discrete Morrey spaces.

2. Discrete Morrey spaces

The classical Morrey spaces $M_{\lambda,p}$, $0 \leq \lambda \leq \frac{1}{p}$, $1 \leq p < \infty$ (see [1, 11, 19, 21, 23, 24, 25]), consist of the functions $f \in L_{p,loc}$ for which the following norm is finite

$$\|f\|_{M_{\lambda,p}} = \sup_{x \in R^d} \sup_{r > 0} \left[|B(x;r)|^{-\lambda} \, \|f\|_{L_p(B(x;r))} \right].$$

We note that if $\lambda = 0$, then $M_{\lambda,p} = L_p$; if $\lambda = \frac{1}{p}$, then $M_{\lambda,p} = L_\infty$ (see [1]). In case $p > 1$, $0 \leq \lambda < \frac{1}{p}$, F.Chiarenza and M.Frasca [7] showed the boundedness of the Hardy–Littlewood maximal operator, the fractional integral operator and a singular integral operator in the Morrey spaces. Hence, in particular, it implies the boundedness of the Riesz transform in Morrey spaces. It means that, in case $p > 1$, $0 \leq \lambda < \frac{1}{p}$, for any $f \in M_{\lambda,p}$ we have $R_j f \in M_{\lambda,p}$, and there exist $C_{\lambda,p} > 0$ such that

$$\|R_j f\|_{M_{\lambda,p}} \leq C_{\lambda,p} \cdot \|f\|_{M_{\lambda,p}}$$

holds for all $f \in M_{\lambda,p}$.

In [12], the authors introduced a discrete analogue of Morrey spaces and studied their inclusion properties. For $m \in Z^d$ and $n \in N \cup \{0\}$ define $S_{m,n} = \{k \in Z^d : \|k - m\| \leq n\}$, where $\|k\| := \max\{|k_1|, ..., |k_d|\}$. Following standard conventions, we denote the cardinality of a set S by $|S|$. Then we have $|S_{m,n}| = (2n+1)^d$ for all $m \in Z^d$ and each

$n \in N \cup \{0\}$. Discrete Morrey spaces $m_{\lambda,p}$, $0 \le \lambda \le \frac{1}{p}$, $1 \le p < \infty$, consist of the sequences $h = \{h_n\}_{n \in Z^d}$ for which the following norm is finite

$$\|h\|_{m_{\lambda,p}} = \sup_{m \in Z^d} \sup_{n \in N \cup \{0\}} \left[|S_{m,n}|^{-\lambda} \left(\sum_{k \in S_{m,n}} |h_k|^p \right)^{1/p} \right].$$

In [2, 3, 13, 14, 15, 16] it was proved the boundedness of the discrete Hardy-Littlewood maximal operators, discrete Riesz potentials, discrete Hilbert transform and discrete Ahlfors-Beurling transforms on discrete Morrey spaces.

Note that if $h = \{h_n\}_{n \in Z^d} \in m_{\lambda,p}$, $1 \le p < \infty$, $0 \le \lambda < \frac{1}{p}$, then the series $\sum_{m \in Z^d,\, m \ne n} \frac{n_j - m_j}{|n-m|^{d+1}} \cdot h_m$ absolutely converges. Indeed, for each $n \in Z^d$ we have

$$\sum_{m \in Z^d,\, m \ne n} \frac{|n_j - m_j|}{|n-m|^{d+1}} \cdot |h_m| \le \sum_{j=1}^{\infty} \sum_{2^{j-1} \le \|n-m\| < 2^j} \frac{|h_m|}{|n-m|^d}$$

$$\le \sum_{j=1}^{\infty} \frac{1}{2^{d(j-1)}} \sum_{\|n-m\| < 2^j} |h_m| \le \sum_{j=1}^{\infty} \frac{1}{2^{d(j-1)}} \left(\sum_{\|n-m\| < 2^j} |h_m|^p \right)^{1/p} \cdot (2^{j+1} - 1)^{d-d/p}$$

$$\le 4^d \|h\|_{m_{\lambda,p}} \cdot \sum_{j=1}^{\infty} 2^{d(j+1)(\lambda - 1/p)} \le \frac{16\|h\|_{m_{\lambda,p}}}{1 - 2^{d(\lambda - 1/p)}}. \tag{1}$$

It follows that if $h = \{h_n\}_{n \in Z^d} \in m_{\lambda,p}$, $1 \le p < \infty$, $0 \le \lambda < \frac{1}{p}$, then the Riesz transform of the sequence h exists.

3. Boundedness of the discrete Riesz transform on discrete Morrey spaces

We next present the main results of this paper.

Theorem 1. *Let $1 < p < \infty$, $0 \le \lambda < 1/p$. For any $h \in m_{\lambda,p}$ we have $\widetilde{R}_j(h) \in m_{\lambda,p}$, and there exists $c_{\lambda,p} > 0$ such that*

$$\|\widetilde{R}_j(h)\|_{m_{\lambda,p}} \le c_{\lambda,p} \cdot \|h\|_{m_{\lambda,p}}$$

holds for all $h \in m_{\lambda,p}$.

Proof. We define the function $f(x)$ to be $\frac{2^d}{\gamma(d)} h_n$ for $x \in P(n, 1/4)$, $n \in Z^d$, and 0 elsewhere, where

$$P(n, \delta) := \{y \in R^d : \|y - n\| < \delta\}.$$

We first show that $f \in M_{\lambda,p}$. Indeed, for any $x \in P(n, 1/2)$, $n \in Z^d$ we have:

if $r \in (0, 1/4]$, then

$$|B(x;r)|^{-\lambda}\|f\|_{L_p(B(x;r))} = |\theta_{(d)} r^d|^{-\lambda} \left(\int_{B(x;r)} |f(y)|^p dy \right)^{1/p}$$

$$\leq |\theta_{(d)} r^d|^{1/p-\lambda} \cdot \frac{2^d}{\gamma_{(d)}} |h_n| \leq \frac{2^d \theta_{(d)}^{1/p-\lambda}}{\gamma_{(d)}} \|h\|_{m_{\lambda,p}}, \tag{2}$$

where $\theta_{(d)} = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$;

if $r \in (1/4, 1]$, then

$$|B(x;r)|^{-\lambda}\|f\|_{L_p(B(x;r))} \leq |\theta_{(d)} r^d|^{-\lambda} \left(\frac{1}{2^d} \sum_{k \in S_{n,1}} |\frac{2^d}{\gamma_{(d)}} h_k|^p \right)^{1/p}$$

$$\leq |\theta_{(d)} r^d|^{-\lambda} \cdot 2^{d-d/p} \cdot \frac{3^{d\lambda}}{\gamma_{(d)}} \|h\|_{m_{\lambda,p}} \leq \frac{2^{d-d/p} 3^{d\lambda}}{\theta_{(d)}^{\lambda} \gamma_{(d)}} \|h\|_{m_{\lambda,p}}; \tag{3}$$

if $r \in (m, m+1]$, $m \in N$, then

$$|B(x;r)|^{-\lambda}\|f\|_{L_p(B(x;r))} \leq |\theta_{(d)} r^d|^{-\lambda} \left(\frac{1}{2^d} \sum_{k \in S_{n,m+1}} \left| \frac{2^d}{\gamma_{(d)}} h_k \right|^p \right)^{1/p}$$

$$\leq |\theta_{(d)} r^d|^{-\lambda} \cdot 2^{d-d/p} \cdot \frac{(2m+3)^{d\lambda}}{\gamma_{(d)}} \|h\|_{m_{\lambda,p}} \leq \frac{2^{d-d/p} 5^{d\lambda}}{\theta_{(d)}^{\lambda} \gamma_{(d)}} \|h\|_{m_{\lambda,p}}. \tag{4}$$

From inequalities (2), (3), (4) it follows that $f \in M_{\lambda,p}$, and

$$\|f\|_{M_{\lambda,p}} = \sup_{x \in R^d} \sup_{r>0} \left[|B(x;r)|^{-\lambda} \|f\|_{L_p(B(x;r))} \right] \leq c_0 \cdot \|h\|_{m_{\lambda,p}},$$

where

$$c_0 = \max \left\{ \frac{2^{d-d/p} 5^{d\lambda}}{\theta_{(d)}^{\lambda} \gamma_{(d)}}, \frac{2^d \theta_{(d)}^{1/p-\lambda}}{\gamma_{(d)}} \right\}.$$

Therefore, $R_j f \in M_{\lambda,p}$ and

$$\|R_j f\|_{M_{\lambda,p}} \leq c_0 \cdot C_{\lambda,p} \|h\|_{m_{\lambda,p}}. \tag{5}$$

Define the function $F(x)$ to be $(\widetilde{R}_j h)_n$ for $x \in P(n, 1/2)$, $n \in Z^d$ and

$$G(x) = (R_j f)(x) - F(x). \tag{6}$$

We first show that $G \in M_{\lambda,p}$. For any $x \in P(n,1/2)$, $n \in Z^d$ we have

$$G(x) = \sum_{m\in Z^d,\, m\neq n} 2^d h_m \int_{P(m,1/4)} \frac{x_j - y_j}{|x-y|^{d+1}} dy - \sum_{m\in Z^d,\, m\neq n} \frac{n_j - m_j}{|n-m|^{d+1}} \cdot h_m$$

$$+2^d h_n \cdot \text{v.p.} \int_{P(n,1/4)} \frac{x_j - y_j}{|x-y|^{d+1}} dy$$

$$= 2^d \sum_{m\in Z^d,\, m\neq n} h_m \int_{P(m,1/4)} \left(\frac{x_j - y_j}{|x-y|^{d+1}} - \frac{n_j - m_j}{|n-m|^{d+1}} \right) dy$$

$$+2^d h_n \cdot \text{v.p.} \int_{P(n,1/4)} \frac{x_j - y_j}{|x-y|^{d+1}} dy = G_1(x) + G_2(x). \tag{7}$$

Let $m \neq n$. Since for every $x \in P(n,1/2)$ and $y \in P(m,1/4)$

$$|n-m| - 3/4 \leq |x-y| \leq |n-m| + 3\sqrt{d}/4, \;\; |n_j - y_j| \leq |n-m| + 3/4,$$

then we get

$$\left| \frac{x_j - y_j}{|x-y|^{d+1}} - \frac{n_j - m_j}{|n-m|^{d+1}} \right| \leq \frac{|x_j - n_j|}{|x-y|^{d+1}}$$

$$+|n_j - y_j| \cdot \left| \frac{1}{|x-y|^{d+1}} - \frac{1}{|n-m|^{d+1}} \right| + \frac{|y_j - m_j|}{|n-m|^{d+1}} \leq \frac{c_1}{|n-m|^{d+1}},$$

where c_1 is a constant depending only d.

Therefore, for every $x \in P(n,1/2)$

$$|G_1(x)| \leq 2^d \sum_{m\in Z^d,\, m\neq n} |h_m| \int_{P(m,1/4)} \left| \frac{x_j - y_j}{|x-y|^{d+1}} - \frac{n_j - m_j}{|n-m|^{d+1}} \right| dy$$

$$\leq \sum_{m\in Z^d,\, m\neq n} \frac{c_1 |h_m|}{|n-m|^{d+1}}. \tag{8}$$

From (8) and (1) it follows that

$$|G_1(x)| \leq \sum_{m\in Z^d,\, m\neq n} \frac{c_1 |h_m|}{|n-m|^{d+1}} \leq c_2 \|h\|_{m_{\lambda,p}}, \tag{9}$$

where $c_2 = \frac{16 c_1}{1 - 2^{d(\lambda - 1/p)}}$.

If $r \in (0,1)$, then we have from (9)

$$|B(x;r)|^{-\lambda} \|G_1\|_{L_p(B(x;r))} = |\theta_{(d)} r^d|^{-\lambda} \left(\int_{B(x;r)} |G_1(y)|^p dy \right)^{1/p}$$

$$\leq c_2 |\theta_{(d)} r^d|^{1/p - \lambda} \|h\|_{m_{\lambda,p}} \leq c_3 \|h\|_{m_{\lambda,p}}, \tag{10}$$

where $c_3 = c_2 \cdot \theta_{(d)}^{1/p-\lambda}$;

if $r \in [k-1, k)$, $k \in N$, $k \geq 2$, then from (8) and from the Hölder's inequality we have

$$|B(x;r)|^{-\lambda} \|G_1\|_{L_p(B(x;r))} = |\theta_{(d)} r^d|^{-\lambda} \left(\int_{B(x;r)} |G_1(y)|^p dy \right)^{1/p}$$

$$\leq |\theta_{(d)} r^d|^{-\lambda} \left(\int_{P(n;k+1/2)} |G_1(y)|^p dy \right)^{1/p}$$

$$= |\theta_{(d)} r^d|^{-\lambda} \left(\sum_{s \in Z^d:\ \|s-n\| \leq k} \int_{P(s;1/2)} |G_1(y)|^p dy \right)^{1/p}$$

$$\leq |\theta_{(d)} r^d|^{-\lambda} \left(\sum_{s \in Z^d:\ \|s-n\| \leq k} \left(\sum_{m \in Z^d,\ m \neq s} \frac{c_1 |h_m|}{|s-m|^{d+1}} \right)^p \right)^{1/p}$$

$$\leq c_1 |\theta_{(d)} r^d|^{-\lambda} \left(\sum_{s \in Z^d:\ \|s-n\| \leq k} \left(\sum_{m \neq s} \frac{|h_m|^p}{|s-m|^{d+1}} \right) \cdot \left(\sum_{m \neq s} \frac{1}{|s-m|^{d+1}} \right)^{p-1} \right)^{1/p}$$

$$\leq c_1 c_4^{1-1/p} |\theta_{(d)} r^d|^{-\lambda} \left(\sum_{s \in Z^d:\ \|s-n\| \leq k} \sum_{m \neq s} \frac{|h_m|^p}{|s-m|^{d+1}} \right)^{1/p}$$

$$= c_1 c_4^{1-1/p} |\theta_{(d)} r^d|^{-\lambda} \left(\sum_{m \in Z^d} |h_m|^p \sum_{\|s-n\| \leq k, s \neq m} \frac{1}{|s-m|^{d+1}} \right)^{1/p}$$

$$\leq \frac{c_1 c_4^{1-1/p}}{|\theta_{(d)} r^d|^{\lambda}} \left(\sum_{\|m-n\| \leq 2k} |h_m|^p c_4 + \sum_{i=1}^{\infty} \sum_{2^i k < \|n-m\| \leq 2^{i+1} k} \frac{|h_m|^p}{k \cdot 2^{i(d+1)-3d-1}} \right)^{1/p}$$

$$\leq \frac{c_1 c_4^{1-1/p}}{|\theta_{(d)} r^d|^{\lambda}} \left(c_4 (2k+1)^{dp\lambda} \|h\|_{m_{\lambda,p}}^p + \sum_{i=1}^{\infty} \frac{(2^{i+1} k)^{dp\lambda} \|h\|_{m_{\lambda,p}}^p}{k \cdot 2^{i(d+1)-3d-1}} \right)^{1/p}$$

$$\leq \frac{c_1 c_4^{1-1/p}}{\theta_{(d)}^{\lambda}} \|h\|_{m_{\lambda,p}} \left(c_4 \cdot 5^{dp\lambda} + \frac{2^{2dp\lambda+3d+1}}{k \cdot (2^{d+1-dp\lambda} - 1)} \right)^{1/p} \leq c_5 \|h\|_{m_{\lambda,p}}, \tag{11}$$

where

$$c_4 = \sum_{m \in Z^d: m \neq 0} \frac{1}{|m|^{d+1}} \leq \sum_{n \in N} \frac{(2n+1)^d - (2n-1)^d}{n^{d+1}} \leq d \cdot 3^{d-2} \pi^2,$$

$$c_5 = \frac{c_1 c_4^{1-1/p}}{\theta_{(d)}^{\lambda}} \left(c_4 \cdot 5^{dp\lambda} + \frac{2^{2dp\lambda+3d+1}}{k \cdot (2^{d+1-dp\lambda} - 1)} \right)^{1/p}.$$

It follows from (10), (11) that $G_1 \in M_{\lambda,p}$ and

$$\|G_1\|_{M_{\lambda,p}} = \sup_{x\in R^d} \sup_{r>0} \left[|B(x;r)|^{-\lambda}\|G_1\|_{L_p(B(x;r))}\right] \le c_6\|h\|_{m_{\lambda,p}}, \tag{12}$$

where $c_6 = \max\{c_3, c_5\}$.

Let us show that $G_2 \in M_{\lambda,p}$.

For every $n \in Z^d$ and $a > 0$ denote

$$\Gamma(n;a) = \{y \in R^d : \|y-n\| = a\},$$

and for every $x \in P(n,1/2)$ denote

$$\delta_x = \rho(x;\Gamma(n;1/4)) = \inf\{|x-y| : y \in \Gamma(n;1/4)\}.$$

For every $x \in P(n,1/2) \setminus \Gamma(n;1/4)$ we have

$$|G_2(x)| = 2^d|h_n| \left|\int_{P(n,1/4)} \frac{x_j-y_j}{|x-y|^{d+1}}dy\right|$$

$$\le 2^d|h_n| \int_{B(x,d)\setminus B(x;\delta_x)} \frac{dy}{|x-y|^d} = 2^d d\theta_{(d)}|h_n| \ln\frac{d}{\delta_x} \tag{13}$$

Since

$$\int_{P(n,1/2)} \left|\ln\frac{d}{\delta_x}\right|^p dx \le 4^d d\int_0^{1/4} \left|\ln\frac{d}{t}\right|^p dt,$$

then for every $n \in Z^d$

$$\int_{P(n,1/2)} |G_2(x)|^p\,dx \le (2^d d\theta_{(d)})^p|h_n|^p \int_{P(n,1/2)} \left|\ln\frac{d}{\delta_x}\right|^p dx \le c_7|h_n|^p, \tag{14}$$

where

$$c_7 = 4^d d\cdot(2^d d\theta_{(d)})^p \int_0^{1/4} \left|\ln\frac{d}{t}\right|^p dt.$$

Let $x \in P(n,1/2)$. If $r \in (0,1)$, then it follows from (13) that

$$|B(x;r)|^{-\lambda}\|G_2\|_{L_p(B(x;r))} = |\theta_{(d)}r^d|^{-\lambda}\left(\int_{B(x;r)}|G_2(y)|^p dy\right)^{1/p}$$

$$\le |\theta_{(d)}r^d|^{-\lambda}\cdot 2^d d\theta_{(d)}\cdot\left[\sum_{\|s-n\|\le 1}|h_s|4^d d\cdot r^{d-1}\int_0^{2r}\left|\ln\frac{d}{t}\right|^p dt\right]^{1/p}$$

$$\le (2d+1)|\theta_{(d)}r^d|^{-\lambda}\cdot 2^d d\theta_{(d)}\cdot\|h\|_{m_{\lambda,p}}(4^d d)^{1/p}\left[r^{d-1}\int_0^{2r}\left|\ln\frac{d}{t}\right|^p dt\right]^{1/p} \le c_8\|h\|_{m_{\lambda,p}}, \tag{15}$$

where

$$c_8 = d(2d+1)\theta_{(d)}^{1-\lambda} \cdot 2^d(4^d d)^{1/p} \sup_{0<r<1} r^{-d\lambda}\left[r^{d-1}\int_0^{2r}\left|\ln\frac{d}{t}\right|^p dt\right]^{1/p} < \infty;$$

if $r \in [k-1,k)$, $k \in N$, $k \geq 2$, then it follows from (14) that

$$|B(x;r)|^{-\lambda}\|G_2\|_{L_p(B(x;r))} = |\theta_{(d)}r^d|^{-\lambda}\left(\int_{B(x;r)}|G_2(y)|^p dy\right)^{1/p}$$

$$\leq |\theta_{(d)}r^d|^{-\lambda}\left(\int_{P(n;k+1/2)}|G_2(y)|^p dy\right)^{1/p}$$

$$= |\theta_{(d)}r^d|^{-\lambda}\left(\sum_{s\in Z^d:\,\|s-n\|\leq k}\int_{P(s;1/2)}|G_2(y)|^p dy\right)^{1/p}$$

$$\leq |\theta_{(d)}r^d|^{-\lambda}\left(\sum_{s\in Z^d:\,\|s-n\|\leq k} c_7|h_n|^p\right)^{1/p}$$

$$\leq \theta_{(d)}^{-\lambda}c_7^{1/p}(k-1)^{-d\lambda}(2k+1)^{d\lambda}\|h\|_{m_{\lambda,p}} \leq c_9\|h\|_{m_{\lambda,p}}, \tag{16}$$

where $c_9 = 3^{d\lambda}c_7^{1/p}\theta_{(d)}^{-\lambda}$.

By (15), (16) we find that $G_2 \in M_{\lambda,p}$ and

$$\|G_2\|_{M_{\lambda,p}} = \sup_{x\in R^d}\sup_{r>0}\left[|B(x;r)|^{-\lambda}\|G_2\|_{L_p(B(x;r))}\right] \leq c_{10}\|h\|_{m_{\lambda,p}}, \tag{17}$$

where $c_{10} = \max\{c_8, c_9\}$.

Then it follows from (7), (12) and (17) that $G \in M_{\lambda,p}$ and

$$\|G\|_{M_{\lambda,p}} \leq (c_6 + c_{10})\|h\|_{m_{\lambda,p}}. \tag{18}$$

Since $F(x) = G(x) + (R_j f)(x)$, by (5) and (18) we get that $F \in M_{\lambda,p}$ and

$$\|F\|_{M_{\lambda,p}} \leq (c_0 \cdot C_{\lambda,p} + c_6 + c_{10})\|h\|_{m_{\lambda,p}}.$$

Therefore, for every $m \in Z^d$ and $n \in N \cup \{0\}$ we have

$$|S_{m,n}|^{-\lambda}\left(\sum_{k\in S_{m,n}}|(\widetilde{R}_j h)_k|^p\right)^{1/p} = (2n+1)^{-d\lambda}\left(\int_{P(m;n+1/2)}|F(x)|^p dx\right)^{1/p}$$

$$\leq (2n+1)^{-d\lambda}\|F\|_{L_p(B(m,\sqrt{d}(n+1/2)))} \leq (2n+1)^{-d\lambda}|B(m,\sqrt{d}(n+1/2))|^{\lambda}\|F\|_{M_{\lambda,p}}$$

$$\leq \theta^{\lambda}_{(d)}(c_0 \cdot C_{\lambda,p} + c_6 + c_{10})\|h\|_{m_{\lambda,p}}.$$

It follows that $\widetilde{R}_j(h) \in m_{\lambda,p}$ and

$$\|\widetilde{R}_j(h)\|_{m_{\lambda,p}} \leq \theta^{\lambda}_{(d)}(c_0 \cdot C_{\lambda,p} + c_6 + c_{10})\|h\|_{m_{\lambda,p}}.$$

This completes the proof of Theorem 1. ◄

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On the Basicity of double System of Exponents in the Weighted Lebesgue Space

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Abstract. This paper considers double system of exponentials with linear phase in the weighted space $L_{p,\rho}$ with power weight $\rho(\cdot)$ on the segment $[-\pi,\pi]$. Under certain conditions on the weight function $\rho(\cdot)$ and on the perturbation parameters, the completeness, minimality and basicity of this system in $L_{p,\rho}$ is proved. An explicit expression for the biorthogonal system in the case of minimality is derived and its integral representation is obtained. The obtained results generalize all previously known results in this direction.

Key Words and Phrases: exponential system, basicity, weighted space.

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1. Introduction

Investigation of many partial differential equation by the application of Fourier method reduces to perturbed trigonometric system of sines (or cosines) of the form

$$\{\sin(nt+\alpha(t))\}_{n\in N}, \tag{1}$$

where $\alpha: [0,\pi]\to R$ is some function (N is a set of natural numbers). Similar problems were studied, for example, in the papers [2, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 42]. To justify the Fourier method it is necessary to study the basicity properties (completeness, minimality, basis property, etc.) of these systems in different functional spaces. Complex versions of these systems are perturbed system of exponents of the form

$$\left\{e^{i(nt+\beta(t)sign\, n)}\right\}_{n\in Z}, \tag{2}$$

where $\beta: [-\pi,\pi]\to R$ is some function (Z is the set of integers). Basis properties of the systems (1) and (2) in corresponding spaces are closely linked, in Lebesgue spaces L_p they are well studied by various mathematicians (see, for example [2, 3, 5, 6, 7, 12, 13, 17, 19, 20, 21, 22, 23, 24, 25, 26, 27]). The case $L_\infty = C[-\pi,\pi]$ is treated in [32]. In connection with application to solution of differential equations, the interest in Lebesgues spaces

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$L_{p(\cdot)}$ with variable summability power $p(\cdot)$ and in Morrey spaces $L^{p,\alpha}$ greatly increased in recent years. Problems of approximation in these spaces have also begun to be studied and basicity problems of the systems (1), (2) in $L_{p(\cdot)}$ are studied in [28, 43], and basicity of the classical system of exponents with linear phase in Morrey spaces are studied in [38, 44, 45]. Note that study of basicity properties of the systems (1), (2) in weighted spaces $L_{p,\rho}$ is equivalent to the study of analogous properties of the systems (1), (2) with corresponding degenerate coefficients in the spaces L_p. For this reason, it can be assumed that the study of the basicity of trigonometric systems in weighted Lebesgue spaces takes its origin from the paper of K.Babenko [18]. Later this area was developed in the works [14, 15, 16, 29, 30, 39, 40, 41]. The problem of basicity of the exponential system in the weighted space $L_{p,\rho} \equiv L_{p,\rho}(-\pi,\pi)$, $1 < p < +\infty$, is solved in the paper [31]. Such a condition is a Muckenhaupt condition with respect to the weight function $\rho(\cdot)$:

$$\sup_I \left(\frac{1}{|I|}\int_I \rho(t)\,dt\right)\left(\frac{1}{|I|}\int_I \rho^{-\frac{1}{p-1}}dt\right)^{p-1} < \infty, \tag{3}$$

where supis taken over all intervals $I \subset [-\pi,\pi]$ and $|I|$ is the length of the interval I.

In the papers [3, 15] the system (2) is considered in the case when $\beta(t) = \beta\, t$, where $\beta \in R$ is some real parameter and its basicity in $L_{p,\rho}$, $1 < p < +\infty$, is studied when $\rho(\cdot)$ has the following form

$$\rho(t) = \prod_{k=-r}^{r} |t - t_k|^{\alpha_k},$$

where $-\pi = t_{-r} < t_{-r+1} < ... < t_r < \pi$.

The class of weights, satisfying the condition (3), is denoted by A_p. It is easy to see that

$$\rho \in A_p \Leftrightarrow -1 < \alpha_k < p-1, \;\; k = \overline{-r,r}.$$

In this paper the minimality of the exponential system

$$\left\{e^{i\left(n+\frac{\beta}{2}sign\, n\right)t}\right\}_{n\in Z},$$

in the weighted space $L_{p,\rho}$, $1 < p < +\infty$, where $\beta \in C$ is a complex parameter, is studied. In contrast to the paper [3], an explicit expression of the biorthogonal system is built and its integral representation is obtained.

2. Preliminaries. Main lemma

Consider the following double system of exponents

$$\left\{e^{i[(n+\beta_1)t+\gamma]};\, e^{-i[(k+\beta_2)t+\gamma_2]}\right\}_{n\in Z_+;k\in N}, \tag{4}$$

where $\beta_k = Re\beta_k + iIm\beta_k$, $\gamma_k = Re\,\gamma_k + iIm\,\gamma_k$, $k = 1,2$, are complex parameters, $Z_+ = \{0\}\bigcup N$. We assume that the weight function $\rho(\cdot)$ is of the following power form

$$\rho(t) = \prod_{k=-r}^{r} |t - t_k|^{\alpha_k},$$

where $-\pi = t_{-r} < t_{-r+1} < ... < t_0 = 0 < ... < t_r < \pi$, $\{\alpha_k\}_{k=\overline{-r,r}} \subset R$ are some numbers. We consider the weighted space $L_{p,\rho}$, $1 < p < +\infty$, with the norm $\|\cdot\|_{p,\rho}$:

$$\|f\|_{p,\rho} = \left(\int_{-\pi}^{\pi} |f(t)|^p \rho(t)\, dt \right)^{1/p}.$$

It is easy to see that basicity properties of the system (4) in $L_{p,\rho}$ are equivalent to basicity properties of the system

$$\left\{ e^{i(n+\beta_1)t}; e^{-i(k+\beta_2)t} \right\}_{n \in Z_+; k \in N}, \tag{5}$$

in $L_{p,\rho}$. We put $g(t) = e^{\frac{i}{2}(\beta_2 - \beta_1)t}$. It is evident that $\exists \delta > 0$:

$$0 < \delta \le |g(t)| \le \delta^{-1} < +\infty, \ \forall t \in [-\pi, \pi].$$

Multiplying the system (5) to the function $g(t)$, we immediately obtain from here that the basicity properties of the system (5) on $L_{p,\rho}$ are equivalent to the basicity properties of the following system

$$\left\{ e^{i\left(n + \frac{\beta}{2} sign\, n\right)t} \right\}_{n \in Z}, \tag{6}$$

on $L_{p,\rho}$, $\beta = \beta_1 + \beta_2$. Thus, the study of basicity properties of the system (4) on $L_{p,\rho}$ is reduced to the investigation of corresponding properties with respect to the system (6) on $L_{p,\rho}$.

Let $\beta \in C$ be some complex number. We will assume throughout the paper that $(1+z)^\beta$ is some fixed branch of multivalued analytic function $(1+z)^\beta$ on the complex plane with the cut along the semiline $(-\infty, -1) \subset R$ on the real axis and take

$$(1+z)^{-\beta} = \frac{1}{(1+z)^\beta}.$$

Analogously, we define a branch z^β of a multivalued function z^β on C with the cut along $(-\infty, 0) \subset R$ and $z^{-\beta} = \frac{1}{z^\beta}$.

We will essentially use the following main lemma in the proof of main results.

Lemma 1. *Let $Re\beta > -1$. Then the following Cauchy integral formulas hold*

$$J_m^-(z) \equiv \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{-i(\beta+m)\theta} \left(1 + e^{i\theta}\right)^\beta}{e^{i\theta} - z} d\theta \equiv \begin{cases} 0, & |z| < 1, \\ -z^{-m-\beta-1}(1+z)^\beta, & |z| > 1, \end{cases}$$

$$J_m^+(z) \equiv \frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{e^{i(m+1)\theta}\left(1+e^{i\theta}\right)^{\beta}}{e^{i\theta}-z}d\theta \equiv \begin{cases} 0, & |z|>1\,, \\ z^m(1+z)^{\beta}\,, & |z|<1\,, \end{cases}$$

$$\forall m \in Z_+.$$

Proof. Consider an expression $J_m^+(z)$. Make the change of variables

$$tg\frac{\theta}{2} = t \Rightarrow e^{i\theta} = \frac{1+it}{1-it}.$$

We have

$$J_m^+(z) = \int_{-\infty}^{+\infty}\frac{(1+it)^{m+1}(1-it)^{-m-1}}{\frac{2^{-\beta}}{(1-it)^{-\beta}}\left(\frac{1+it}{1-it}-z\right)}\frac{2dt}{(1-it)(1+it)} =$$

$$= 2^{\beta+1}\int_{-\infty}^{+\infty}(1-it)^{-\beta}(1-it)^{-m-1}(1+it)^m[1+it-z+izt]^{-1}dt =$$

$$= \frac{2^{\beta+1}}{i(1+z)}\int_{-\infty}^{+\infty}(1-it)^{-\beta}(1-it)^{-m-1}\left[t-\frac{i(1-z)}{1+z}\right]^{-1}(1+it)^m dt.$$

Let $Rez = x,\ \ Imz = y$. We obtain

$$Im\left(i\frac{1-z}{1+z}\right) = \frac{1-x^2-y^2}{(1+x)^2+y^2} > 0, \qquad |z|<1,$$

and it is evident that

$$Im\left(i\frac{1-z}{1+z}\right) < 0, \qquad |z|>1.$$

Denote the integrand function by $F(w)\,,\ \ w \in C$:

$$F(w) = (1-iw)^{-\beta-m-1}(1+iw)^m\left(w-i\frac{1-z}{1+z}\right)^{-1}.$$

It is obvious that for large values of $|w|$ the following estimation holds

$$|F(w)| \le \frac{M}{|w|^{2+Re\beta}},$$

where $M>0$ is some constant. Applying Theorem 5.3 from monograph [1] (see p. 127), we obtain that

$$J_m^+(z) = \frac{2^{\beta+1}}{i(1+z)}2\pi i \underset{t=i\frac{1-z}{1+z}}{Res}\left[(1-it)^{-\beta-m-1}(1+it)^m\left(t-i\frac{1-z}{1+z}\right)^{-1}\right] =$$

$$= \frac{2^{\beta+2}\pi}{1+z}\left(\frac{2}{1+z}\right)^{-\beta-m-1}\left(\frac{2z}{1+z}\right)^m = 2\pi z^m(1+z)^{\beta}\,, \ \ |z|<1,$$

since for $|z| < 1$, the only pole of the function $F(w)$ in the upper half-plane is $w = i\frac{1-z}{1+z}$. Analogous reasoning implies that $J_m^+(z) \equiv 0,\ \ |z| > 1$, since for $|z| > 1$ the function $F(w)$ has no poles in the upper half-plane.

The formula for $J_m^-(z)$ is proved in a similar way.

The lemma is proved.

3. Minimality in $L_{p,\rho}$

Consider the following system of functions

$$\vartheta_n^+(t) = \frac{e^{-i\frac{\beta}{2}t}}{2\pi}\left(1+e^{it}\right)^{\beta}\sum_{k=0}^{n} C_{-\beta}^{n-k}e^{-ikt},\ \ n \in Z_+;$$
$$\vartheta_m^-(t) = -\frac{e^{-i\frac{\beta}{2}t}}{2\pi}\left(1+e^{it}\right)^{\beta}\sum_{k=1}^{m} C_{-\beta}^{m-k}e^{ikt},\ \ m \in N;$$

where

$$C_{-\gamma}^k = \frac{\gamma(\gamma-1)\ldots(\gamma-k+1)}{k!},$$

is a binomial coefficient. Accordingly, we denote

$$e_n^+(t) \equiv e^{i\left(n+\frac{\beta}{2}\right)t},\ \ n \in Z_+;\ \ e_k^-(t) \equiv e^{-i\left(n+\frac{\beta}{2}\right)t},\ \ k \in N.$$

Assume that $Re\beta > -1$. The expansion in powers of z of the function $(1+z)^{-\beta}J_m^+(z)$ that is analytic on $|z| < 1$ is

$$(1+z)^{-\beta}J_m^+(z) = \sum_{n=0}^{\infty} a_{n;m}^+ z^n,$$

where

$$a_{n;m}^+ = \int_{-\pi}^{\pi} e^{i\left(m+\frac{\beta}{2}\right)t}\vartheta_n^+(t)\,dt.$$

On the other hand, it follows from Lemma 1 that

$$(1+z)^{-\beta}J_m^+(z) \equiv z^m,\quad |z| < 1.$$

Comparing the corresponding coefficients, we arrive at the following equalities

$$\int_{-\pi}^{\pi} e_m^+(t)\,\vartheta_n^+(t)\,dt = \delta_{nm},\quad \forall n, m \in Z_+.$$

Expanding the function $(1+z)^{-\beta}J_m^+(z)$ at infinity in powers of z^{-1}, we obtain

$$(1+z)^{-\beta}J_m^+(z) = \sum_{n=1}^{\infty} b_{n;m}^+ z^{-n},\ \ |z| > 1,$$

where

$$b_{n;m}^{+}=\int_{-\pi}^{\pi}e^{i\left(m+\frac{\beta}{2}\right)t}\vartheta_{n}^{-}\left(t\right)dt,\quad m\in Z_{+},\quad n\in N.$$

It is easy to see that

$$\lim_{|z|\to\infty}\left(1+z\right)^{-\beta}J_{m}^{+}\left(z\right)=0.$$

On the other hand, again, as follows from Lemma 1, we have

$$\left(1+z\right)^{-\beta}J_{m}^{+}\left(z\right)\equiv 0,\quad |z|>1.$$

These two expansions imply

$$\int_{-\pi}^{\pi}e^{i\left(m+\frac{\beta}{2}\right)t}\vartheta_{n}^{-}\left(t\right)dt=0,\quad \forall m\in Z_{+},\quad \forall n\in N.$$

The relations

$$\begin{array}{l}\int_{-\pi}^{\pi}e_{m}^{-}\left(t\right)\vartheta_{n}^{+}\left(t\right)dt=0,\quad m\in N,\quad n\in Z_{+};\\ \int_{-\pi}^{\pi}e_{m}^{-}\left(t\right)\vartheta_{n}^{-}\left(t\right)dt=\delta_{nm},\qquad \forall n,m\in N\end{array}$$

can be proved analogously. As a result, we obtain the validity of the following statement.

Proposition 1. *Let $Re\beta>-1$. Then for all admissible values of indices n and m the following relations*

$$\int_{-\pi}^{\pi}e_{n}^{\pm}\left(t\right)\vartheta_{m}^{\pm}\left(t\right)dt=\delta_{nm},\qquad \int_{-\pi}^{\pi}e_{n}^{\pm}\left(t\right)\vartheta_{m}^{\mp}\left(t\right)dt=0,$$

hold.

Consider the following proposition.

Proposition 2. *Let $1<p<+\infty$ and $\frac{1}{p}+\frac{1}{q}=1$. Then $\left(L_{p,\rho}\right)^{*}=L_{q,\rho}$ and every functional $\vartheta^{*}\in\left(L_{p,\rho}\right)^{*}$ is represented, by the uniquely determined for it function $\vartheta\in L_{q,\rho}$, by the following expression*

$$\vartheta^{*}\left(f\right)=\int_{-\pi}^{\pi}f\bar{\vartheta}\rho dt,\quad \forall f\in L_{p,\rho}.$$

Following this proposition, define the following system of functions

$$h_{n}^{\pm}\left(t\right)=\rho^{-1}\left(t\right)\overline{\vartheta_{n}^{\pm}\left(t\right)}.$$

It is easy to see that the system $\{h_{n}^{\pm}\}$ belongs to the space $L_{q,\rho}$ when

$$\alpha_{k}<\frac{1}{q-1},\;\; k=\overline{-r+1,r-1};\;\; Re\beta-\frac{\alpha_{\pm r}}{p}>-\frac{1}{q}.$$

This follows directly from the representation of $\{\vartheta_{n}^{\pm}\}$ and from the relation

$$\int_{-\pi}^{\pi}\left|h_{n}^{\pm}\right|^{q}\rho dt=\int_{-\pi}^{\pi}\rho^{1-q}\left|\vartheta_{n}^{\pm}\right|^{q}dt.$$

Taking into account that $\frac{1}{q-1}=\frac{p}{q}$, we obtain the following theorem from Proposition 1 and 2.

Theorem 1. *Assume that the following inequalities hold*

$$Re\beta > -1; \quad -1 < \alpha_k < \tfrac{p}{q}, \quad k = \overline{-r+1, r-1};$$

$$-1 < \alpha_{\pm r} < \tfrac{p}{q} + pRe\beta.$$

Then the exponential system $\left\{ e^{i\left(n + \frac{\beta}{2} sign\, n\right)t} \right\}_{n \in Z}$ *is minimal in* $L_{p,\rho}$, $1 < p < +\infty$.

The following lemma plays a very important role in the study of orthogonal series.

Lemma 2. *The system* $\{\vartheta_n^{\pm}\}$ *has the following integral representation*

$$\vartheta_{n-1}^{+}(t) = \frac{e^{-i\left(n+\frac{\beta}{2}\right)t}}{2\pi}\left(1+e^{it}\right)^{\beta}\left[\left(1+e^{it}\right)^{-\beta} - \frac{\sin \pi\beta}{\pi} e^{i(t+\pi)n} \int_0^1 \frac{x^{n+\beta-1}}{1+xe^{it}} dx\right],$$

$$\vartheta_{n}^{-}(t) = \frac{e^{i\left(n-\frac{\beta}{2}\right)t}}{2\pi}\left(1+e^{it}\right)^{\beta}\left[\left(1+e^{-it}\right)^{-\beta} - \frac{\sin \pi\beta}{\pi} e^{i(\pi-t)n} \int_0^1 \frac{x^{n+\beta-1}(1-x)^{-\beta}}{1+xe^{it}} dx\right], \quad n \in N,$$

for $Re\beta \in (-1, 1)$.

Proof. We will prove this lemma with regard to ϑ_n^- since for ϑ_n^+ it is proved in exactly the same way. Thus, let

$$\vartheta_n^-(t) = -\frac{e^{-i\frac{\beta}{2}t}}{2\pi}\left(1+e^{it}\right)^{\beta} \sum_{k=1}^{n} C_{-\beta}^{n-k} e^{itk}.$$

Make the following transformation

$$\sum_{k=1}^{n} C_{-\beta}^{n-k} e^{ikt} = e^{int} \sum_{k=0}^{n-1} C_{-\beta}^{k} e^{-ikt} = e^{int}\left[\left(1+e^{-it}\right)^{-\beta} - \sum_{k=n}^{\infty} C_{-\beta}^{k} e^{-ikt}\right] =$$

$$= e^{int}\left[\left(1+e^{-it}\right)^{-\beta} - e^{-int} \sum_{k=0}^{\infty} C_{-\beta}^{k+n} e^{-ikt}\right] =$$

$$= e^{int}\left[\left(1+e^{-it}\right)^{-\beta} - e^{-int} \frac{(-1)^n (\beta)_n}{n!} F\left(1; n+\beta; n+1; e^{i(\pi-t)}\right)\right],$$

where

$$(\beta)_n = \beta(\beta+1)\ldots\ldots(\beta+n-1) = \frac{\Gamma(\beta+n)}{\Gamma(\beta)},$$

$\Gamma(\cdot)$ is Euler's gamma function and $F(a;b;c;z)$ is hypergeometric function.

Using integral representation for hypergeometric functions (see [46], p. 72), we find from here that

$$\sum_{k=1}^{n} C_{-\beta}^{n-k} e^{ikt} = e^{int}\left[\left(1+e^{-it}\right)^{-\beta} - e^{i(\pi-t)n}\frac{\sin \pi\beta}{\pi}\int_0^1 \frac{x^{n+\beta-1}(1-x)^{-\beta}}{1+xe^{-it}}dx\right].$$

Substituting this representation into the expression of ϑ_n^-, we arrive at the required fact.

The lemma is proved.

4. Completeness in $L_{p,\rho}$

The following lemma on the uniform convergence plays an important role in the study of the completeness of the exponential system (6) in $L_{p,\rho}$.

Lemma 3. *Let $-1 < Re\beta < 0$ and $\beta = 0$. If $\psi(\cdot)$ is an arbitrary Holder function on $[-\pi,\pi] : e^{i\beta\pi}\psi(-\pi) = \psi(\pi) = 0$, then the series*

$$\sum_{n=0}^{\infty} a_n^+ e_n^+(t) + \sum_{n=1}^{\infty} a_n^- e_n^-(t),$$

uniformly converges to $\psi(\cdot)$ on $[-\pi,\pi]$, where $a_n^{\pm} = \int_{-\pi}^{\pi}\psi(t)\, h_n^{\pm}(t)\,\rho(t)\,dt$.

Proof. Consider the following conjugate problem: find a piecewise analytic function $F(z)$ inside and outside of the unit circle, which the boundary values on the unit circle satisfy the following condition

$$F^+\left(e^{it}\right) + e^{-i\beta t}F^-\left(e^{it}\right) = e^{-i\frac{\beta}{2}t}\psi(t),\ \ t\in(-\pi,\pi]. \tag{7}$$

We will solve this problem by the method developed in the monograph F.D. Gakhov [1, page 427]. Consider the following multi-valued analytic function in the complex plane

$$\omega(z) = (z+1)^{\gamma}.$$

We carry out cut on the plane z from zero to infinity $(-\infty)$ along the negative real axis. In the cut plane like that, this function will be unique, and the incision for it will be a line of discontinuity. Denote this branch by

$$\omega_{-1}(z) = (z+1)_{-1}^{\gamma}.$$

Let us define

$$\gamma = \frac{1}{2\pi i}\ln e^{-i2\beta\pi} \Rightarrow Re\gamma = -Re\beta.$$

A solution of problem (7) is the following Cauchy type integral

$$F^+(z) = (z+1)_{-1}^{\gamma} X_1^+(z)\,\Psi^+(z)\,; F^-(z) = \left(\frac{z+1}{z}\right)_{-1}^{\gamma} X_1^-(z)\,\Psi^-(z),$$

where

$$X_1(z) = \exp[\Gamma(z)],$$

$$\Gamma(z) = \frac{1}{2\pi i}\int_L \frac{\ln[\tau^{-\gamma}G(\arg\tau)]}{\tau - z}d\tau,$$

$$\Psi(z) = \frac{1}{2\pi i}\int_L \frac{(\tau+1)_{-1}^{-\gamma}\varphi(\arg\tau)}{X_1^+(\tau)(\tau - z)}d\tau,$$

$$G(t) = e^{-i\beta t};\ \varphi(t) = e^{-i\frac{\beta}{2}t}\psi(t),$$

$L-$ is a unit circle, which goes around from the point $e^{-i\pi}$ to the point $e^{i\pi}$ in the positive direction.

The fact that $F(z)$ satisfies the boundary condition (7), follows directly from the Sokhotskii-Plemelj formulas. Let $0 < Re\gamma < 1$. It is clear that the function $G(t)$ satisfies the Holder condition on the interval $[-\pi, \pi]$. Moreover, it is easy to verify that the function $\tau^{-\gamma}G(\arg\tau)$ is continuous at a point $\tau = -1$, and as a result it satisfies a certain Holder condition on the unit circle. Then according to the results of the monograph F.D.Gakhov [1, page 55] the function $X_1^{\pm}(\tau)$ satisfies the Holder condition on L. Denote

$$L_{-\pi} = \left\{z = e^{it} : t \in \left[-\pi, -\frac{\pi}{2}\right]\right\}.$$

Assume

$$\varphi^*(\tau) = \frac{\psi(\arg\tau)}{X^+(\tau)\tau^{\frac{\beta}{2}}}, \qquad \tau \in L.$$

Let $\left[(z+1)^{-\gamma}\right]^*$ be a branch, that is holomorphic on the cut along $L_{-\pi}$ on the plane of the function $(z+1)^{-\gamma}$, having the values $(t+1)_{-1}^{-\gamma}$ on the left side of $L_{-\pi}$. So $\left[(t+1)^{-\gamma}\right]^* = (t+1)_{-1}^{-\gamma}$ on $L_\pi = \left\{z = e^{it} : t \in \left[\frac{\pi}{2}, \pi\right]\right\}$,
using the results of the monograph [1] (see page 74), the function $\Psi(z)$ in the vicinity of the point $z = -1$ on the contour L can be represented as

$$\Psi(t) = \left[\frac{e^{i\gamma\pi}}{2i\sin\gamma\pi}\varphi^*(-1+0) - \frac{ctg\,\gamma\pi}{2i}\varphi^*(-1-0)\right]\frac{1}{[(t+1)^\gamma]^*} + \Phi(t), \text{ for } t \in L_\pi; \quad (8)$$

$$\Psi(t) = \left[\frac{ctg\,\gamma\pi}{2i}\varphi^*(-1+0) - \frac{e^{-i\gamma\pi}}{2i\sin\gamma\pi}\varphi^*(-1-0)\right]\frac{1}{[(t+1)^\gamma]^*} + \Phi(t) \text{ for } t \in L_{-\pi}, \quad (9)$$

where under $\left[(t+1)^{-\gamma}\right]^*$ we mean the limit of the function $\left[(z+1)^{-\gamma}\right]^*$, when z tends to t on the left of $L_{-\pi}\bigcup L_\pi$, and moreover

$$\Phi(t) = \frac{\Phi^*(t)}{|t+1|^{\gamma_0}},\ \ \gamma_0 < Re\,\gamma, \quad (10)$$

and the function $\Phi^* (t)$ belongs to the Holder class at the neighborhood of the point $z=-1$.

Applying the Sokhotskii-Plemelj formula from these representations near the point $z=-1$ we have

$$F^+ (t) = (t+1)_{-1}^{\gamma} X_1^+ (t) \left[\frac{1}{2} (t+1)_{-1}^{-\gamma} \varphi^* (t) + \frac{1}{2\pi i} \int_L \frac{\varphi^* (\tau)}{(\tau+1)_{-1}^{\gamma} (\tau - t)} d\tau \right] =$$

$$= X_1^+ (t) \left[\frac{1}{2} \varphi^* (t) + \right.$$

$$\left. + \frac{e^{i\gamma\pi}}{2i \sin \gamma\pi} \varphi^* (-1+0) - \frac{ctg\, \gamma\pi}{2i} \varphi^* (-1-0) \right] + (t+1)_{-1}^{\gamma} X_1^+ (t) \Phi (t) .$$

Passing to the limit as $t \to -1-0$, and taking into account the relation (10), we obtain

$$F^+ (-1-0) = X_1^+ (-1) \left[\frac{1}{2} \varphi^* (-1-0) + \frac{e^{i\gamma\pi}}{2i \sin\, \gamma\pi}\ \varphi^* (-1+0) - \frac{ctg\, \gamma\pi}{2i} \varphi^* (-1-0) \right] =$$

$$= X_1^+ (-1) \left[\frac{e^{i\gamma\pi}}{2i \sin \gamma\pi} \varphi^* (-1+0) - \frac{e^{-i\gamma\pi}}{2i \sin \gamma\pi}\ \varphi^* (-1-0) \right] .$$

Similarly, from the expressions (8) and (9) we obtain

$$F^+ (-1+0) = X_1^+ (-1) \left[\frac{1}{2} \varphi^* (-1+0) + \frac{ctg\gamma\pi}{2i} \varphi^* (-1+0) - \frac{e^{-i\gamma\pi}}{2i \sin \gamma\pi} \varphi^* (-1-0) \right] =$$

$$= X_1^+ (-1) \left[\frac{e^{i\gamma\pi}}{2i \sin \gamma\pi} \varphi^* (-1+0) - \frac{e^{-i\gamma\pi}}{2i \sin \gamma\pi} \varphi^* (-1-0) \right] .$$

Thus, $F^+ (-1-0) = F^+ (-1+0)$, i.e. $F^+ (t)$ is continuous at the point $z=-1$, and as a result, it satisfies a certain Holder condition on L. Expanding $F^+ (z)$ on z at zero, we obtain

$$F^+ (z) = \sum_{n=0}^{\infty} a_n^+ z^n ,$$

where

$$a_n^+ = \int_{-\pi}^{\pi} \psi (t) h_n^+ (t) \rho (t) dt, \quad n \in Z_+ .$$

We have

$$\frac{1}{2\pi i} \int_{|z|=r<1} F^+ (z) z^{-n-1} dz = \begin{cases} a_n^+ , & n \geq 0, \\ 0, & n < 0. \end{cases}$$

Passing to the limit as $r \to 1-0$, hence we get

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} F^+ \left(e^{it} \right) e^{-int} dt = \begin{cases} a_n^+ , & n \geq 0 , \\ 0, & n < 0. \end{cases}$$

As, the function $F^{+}\left(e^{it}\right)$ satisfies Holder condition on L, then its Fourier series on classical system of exponents $\left\{e^{int}\right\}_{n\in Z}$ uniformly converges to it on $[-\pi,\pi]$, and consequently

$$F^{+}\left(e^{it}\right)=\sum_{n=0}^{\infty}a_n^{+}e^{int},\quad t\in[-\pi,\pi].$$

Now, we investigate the boundary properties of the function $F^{-}(z)$. Similarly to the case $F^{+}(z)$, using the representation (8) - (10), and the Sohotskogo- Plemelj formula, we obtain

$$F^{-}(t)=t_{-1}^{-\gamma}(1+t)_{-1}^{\gamma}X_1^{-}(t)\,\Psi^{-}(t)=t_{-1}^{-\gamma}(1+t)_{-1}^{\gamma}X_1^{-}(t)$$
$$\left\{-\frac{1}{2}\varphi^{*}(t)(1+t)_{-1}^{-\gamma}+\left[\frac{e^{i\gamma\pi}}{2i\sin\gamma\pi}\varphi^{*}(-1+0)-\frac{ctg\gamma\pi}{2i}\varphi^{*}(-1-0)\right](1+t)_{-1}^{-\gamma}+\Phi(t)\right\}=$$

$$=t_{-1}^{-\gamma}X_1^{-}(t)\left[-\frac{\varphi^{*}(t)}{2}+\frac{e^{i\gamma\pi}}{2i\sin\gamma\pi}\varphi^{*}(-1+0)-\frac{ctg\gamma\pi}{2i}\varphi^{*}(-1-0)+(1+t)_{-1}^{\gamma}\Phi(t)\right].$$

Passing to the limit as $t\to 1-0$, we have

$$F^{-}(-1-0)=e^{-i\gamma\pi}X_1^{-}(-1)\left[-\frac{\varphi^{*}(-1-0)}{2}+\frac{e^{i\gamma\pi}}{2i\sin\gamma\pi}\varphi^{*}(-1+0)-\frac{e^{i\gamma\pi}}{2i\ \sin\gamma\pi}\varphi^{*}(-1-0)\right]=$$

$$=e^{-i\gamma\pi}X_1^{-}(-1)\frac{e^{i\gamma\pi}}{2i\sin\gamma\pi}\left[\varphi^{*}(-1+0)-\varphi^{*}(-1-0)\right]=0.$$

$$F^{-}(-1+0)=e^{i\gamma\pi}X_1^{-}(-1)\left[-\frac{\varphi^{*}(-1+0)}{2}+\frac{e^{i\gamma\pi}}{2i\sin\gamma\pi}\varphi^{*}(-1+0)-\frac{ctg\gamma\pi}{2i}\varphi^{*}(-1-0)\right]=$$

$$=e^{i\gamma\pi}X_1^{-}(-1)\frac{ctg\gamma\pi}{2i}\left[\varphi^{*}(-1+0)-\varphi^{*}(-1-0)\right]=0.$$

Thus, $F^{-}(-1-0)=F^{-}(-1+0)=0$, and as a result, $F^{-}(t)$ satisfies Holder condition on L. Similarly to the case $F^{+}(t)$, it is proved that the series

$$\sum_{n=1}^{\infty}a_n^{-}e^{-int},$$

uniformly converges to $F^{-}\left(e^{it}\right)$ on $[-\pi,\pi]$. Then from the boundary condition (7) it follows that the biorthogonal series

$$\sum_{n=0}^{\infty}a_n^{+}e_n^{+}(t)+\sum_{n=1}^{\infty}a_n^{-}e_n^{-}(t),$$

uniformly converges to $\psi(t)$ on $[-\pi,\pi]$.

The lemma is proved.

Using the representation (8), (9) and the expression of the functions $F^{\pm}(z)$ we establish the validity of the following lemma.

Lemma 4. *Let $0 < Re\beta < 1$ and $\psi(\cdot)$ be an arbitrary Holder function on $[-\pi,\pi]$. Then the series*

$$\sum_{n=0}^{\infty} a_n^+ e_n^+(t) + \sum_{n=1}^{\infty} a_n^- e_n^-(t),$$

where

$$a_n^{\pm} = \int_{-\pi}^{\pi} \psi(t)\, h_n^{\pm}(t)\, \rho(t)\, dt,$$

uniformly converges to $\psi(\cdot)$ on every compact $G \subset (-\pi,\pi)$, and if $\left|1+e^{it}\right|^{-Re\beta} \in L_{p,\rho}$ converges to it in $L_{p,\rho}$, and the following inequality holds

$$-1 < \alpha_k < \frac{p}{q},\ k = -\overline{r,r}. \tag{11}$$

Indeed, the first part follows from the fact that in this case the functions $F^{\pm}\left(e^{it}\right)$ are Holder functions on each compact $G \subset (-\pi,\pi)$. And under fulfilling the inequality (11) the system of exponents $\left\{e^{int}\right\}_{n\in Z}$ forms a basis for $L_{p,\rho}$ and from the inclusion $\left|1+e^{it}\right|^{-Re\beta} \in L_{p,\rho}$ follows that $F^{\pm}\left(e^{it}\right) \in L_{p,\rho}$.

The following theorem follows directly from these lemmas.

Theorem 2. *Let $\rho \in L_1$ and the parameter β satisfy one of the following conditions:*

i) $-Re\beta \in \bigcup_{k=0}^{\infty}(k,k+1)$;

ii) $-\beta \in Z_+$;

iii) $\left|1+e^{it}\right|^{-Re\beta} \in L_{p,\rho}$ and the following inequalities hold

$$-1 < \alpha_k < \frac{p}{q}, k = -\overline{r,r}.$$

Then the system (3) *is complete in $L_{p,\rho}$, for $\forall p \geq 1$, if $\rho \in L_1$.*

Indeed, let us consider the case *i*), the case *ii*) proves similarly. Let, for example, $-Re\beta \in (1,2)$, i.e. $-2 < Re\beta < -1 \Rightarrow Re\tilde{\beta} < 0$, where $\tilde{\beta} = \beta + 1$. Consider the system

$$\left\{e^{i\left(n+\frac{\beta}{2}\right)t}; e^{-i\left(n+\frac{\beta}{2}\right)t}\right\}_{n\in N}. \tag{12}$$

Presenting this system in the form of

$$\left\{e^{i\left(n-1+1+\frac{\beta}{2}\right)t}; e^{-i\left(n+\frac{\beta}{2}\right)t}\right\}_{n\in N},$$

and multiplying it by $e^{-i\frac{\beta}{2}t}$, we get the following system

$$\left\{e^{i\left(n+\frac{\tilde{\beta}}{2}\right)t}\right\}_{n\in Z}, \tag{13}$$

where $\tilde{\beta} = \beta + 1$, as a result, as it follows from Lemma 3, the corresponding biorthogonal series of an arbitrary Holder function $f \in C^{\alpha}[-\alpha, \pi] : f(-\pi) = f(\pi) = 0$ uniformly converges to it on $[-\pi, \pi]$. Denote the partial sums of this series by $S_m(f)$, $m \in N$. Consequently

$$\|f - S_m(f)\|_{p,\rho}^p = \int_{-\pi}^{\pi} |f(t) - S_m(f)(t)|^p \rho(t)\, dt \leq$$

$$\leq \int_{-\pi}^{\pi} \rho(t)\, dt \max_{[-\pi,\pi]} |f(t) - S_m(f)(t)|^p \to 0,\ m \to \infty.$$

Since the set of such functions is dense in $L_{p,\rho}$, hence, we obtain the completeness of the system (13), and at the same time of the system (12) in $L_{p,\rho}$. From the completeness of the system (12) follows the completeness of the system (6) in $L_{p,\rho}$. The remaining cases are proved by mathematical induction. Case *iii*) directly follows from the Lemma 4.

5. Basicity in $L_{p,\rho}$

Before we proceed to the study of basicity let us mention the conditions for completeness and minimality of the system (6) in $L_{p,\rho}$. Thus, by combining the results of Theorems 1 and 2 we arrive at the following conclusion.

Theorem 3. *Let the weight $\rho(\cdot)$ and parameter β satisfy the following conditions*

$$-1 < \alpha_k < \frac{p}{q},\ k = -\overline{r, r};$$

$$-1 < \alpha_{\pm r} - pRe\beta < \frac{p}{q}.$$

Then the system (6) is complete and minimal in $L_{p,\rho}$, $1 < p < +\infty$.

Now, we turn to the study of basicity of the system (6) in $L_{p,\rho}$. Let all the conditions of Theorem 3 are fulfilled. Let us consider the boundary value problem (7), where $\psi(\cdot)$ is an arbitrary, finite Holder function on $[-\pi, \pi]$. Applying the Sokhotskii-Plemelj formula to expressions $F^{+}(z)$ and $F^{-}(z)$, we obtain

$$F^{+}\left(e^{it}\right) = ie^{-i\frac{\beta}{2}t}\psi(t) + \Phi^{+}(\psi),$$

$$F^{-}\left(e^{it}\right) = ie^{+i\frac{\beta}{2}t}\psi(t) + \Phi^{-}(\psi),$$

where $\Phi^{\pm}(\psi)$ are corresponding Cauchy type integrals, obtained after the application of these formulas

$$\Phi^{+}(\psi)(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{-i\frac{\beta}{2}\theta}\psi(\theta)\, d\theta}{\left(1 + e^{i\theta}\right)^{-\beta}\left(1 - e^{i(t-\theta)}\right)} \left(1 + e^{it}\right)^{-\beta},$$

$$\Phi^{-}(\psi)(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{-i\frac{\beta}{2}\theta}\psi(\theta)\, d\theta}{\left(1 + e^{i\theta}\right)^{-\beta}\left(1 - e^{i(t-\theta)}\right)} \left(1 + e^{-it}\right)^{-\beta}.$$

Consider the biorthogonal series of function $\psi(\cdot)$ on system (6):

$$\psi(t) \sim \sum_{n=0}^{\infty} a_n^+ e_n^+(t) + \sum_{n=1}^{\infty} a_n^- e_n^-(t),$$

and let

$$S_{n,m}(\psi) = S_n^+(\psi) + S_m^-(\psi),\ n \in Z_+,\ m \in N,$$

be its partial sums, where

$$S_n^+(\psi) = \sum_{k=0}^{n} a_k^+ e_k^+(t),\ S_m^-(\psi) = \sum_{k=1}^{m} a_k^- e_k^-(t),$$

and

$$a_k^{\pm} = \int_{-\pi}^{\pi} \psi(t)\, h_k^{\pm}(t)\, \rho(t)\, dt,$$

be corresponding biorthogonal coefficients.

As it has been proven, $S_n^{\pm}$ are the partial sums of biorthogonal series of functions $F^{\pm}\left(e^{it}\right)$, respectively. From the conditions of the theorem and representation of function $F^{\pm}(\cdot)$ it follows that $F^{\pm} \in L_{p,\rho}$. So, under the conditions (11), the system of exponents $\left\{e^{int}\right\}_{n\in Z}$ forms a basis for $L_{p,\rho}$ (see e.g. [31]), then it is clear that $\exists M_1 > 0:$

$$\left\|S_n^{\pm}(\psi)\right\|_{p,\rho} \leq M_1 \left\|F^{\pm}(\cdot)\right\|_{p,\rho},\ \forall n.$$

Consequently

$$\left\|S_{n,m}(\psi)\right\|_{p,\rho} \leq M_1 \left(\left\|F^+\right\|_{p,\rho} + \left\|F^-\right\|_{p,\rho}\right),\ \forall n, m.$$

We have

$$\left\|F^{\pm}\right\|_{p,\rho} \leq \|\psi\|_{p,\rho} + \left\|\Phi^{\pm}(\psi)\right\|_{p,\rho}.$$

On the other hand, from the boundedness of a singular integral operator in weighted space (see e.g. [7, 8]) it follows that under the conditions of the theorem, the following inequality holds

$$\exists M_2 > 0 : \left\|\Phi^{\pm}(\psi)\right\|_{p,\rho} \leq M_2 \|\psi\|_{p,\rho}.$$

Taking into account these inequalities as a result we get

$$\left\|S_{n,m}(\psi)\right\|_{p,\rho} \leq M \|\psi\|_{p,\rho},\ \forall n, m. \tag{14}$$

Further, should pay attention to the fact that the linear manifold of finite, Holder functions on $[-\pi, \pi]$ are dense in $L_{p,\rho}$. Then, the uniformly boundedness of projectors $\{S_{n;m}\}$ in $L_{p,\rho}$ directly follows from (14). As a result, taking into account that under the conditions of the theorem the system (6) is complete and minimal in $L_{p,\rho}$, from the basicity criterion we obtain its basicity in $L_{p,\rho}$.

Theorem is proved.

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On the Basis Property in L_p of Eigenfunctions of a Differential Operator with Integral Boundary Conditions

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Abstract. This paper studies a second-order differential operator with integral boundary conditions of the form $\int_0^1 \varphi_\nu(x)\, y(x) dx = 0, \quad \nu = 1,2$. Conditions on the functions $\varphi_\nu(x), \nu = 1,2$, are found under which the system of eigenfunctions of the differential operator forms a basis in a certain subspace of $L_p(0,1)$, $1 < p < \infty$, of codimension 2. The question of a possible extension of this system to a basis of the entire space $L_p(0,1)$ is also considered.

Key Words and Phrases: second-order differential operator, eigenfunction, boundary conditions, basis.

2010 Mathematics Subject Classifications: 34B24

1. Introduction and Formulation of Problem

Consider the linear differential expression

$$l(y) = -y'' + q(x)\, y, x \in (0,1), \tag{1}$$

and the boundary conditions

$$U_1(y) = U_2(y) = 0, \tag{2}$$

where $q(x)-$ is a complex-valued integrable function on $[0,1]$ and $U_1(y)$ and $U_2(y)$ - are the corresponding boundary forms. The differential expression (1) and the boundary conditions (2) generate a differential operator L with a domain $\mathrm{D}(L)$ in some functional space X. We are interested in the behavior of the eigenvalues and eigenfunctions of this differential operator. This problem has been well studied in the case of regular boundary conditions $U_\nu(y) = 0$, $\nu = 1,2$, (see [1,2] and the references therein). The case of irregular, as well as more general regular boundary conditions involving certain integrals of the function $y(x)$ and its derivatives, has been considered in [3-7]. These works studied the spectral properties of the corresponding operator (spectrality, eigenvalues, and eigenfunctions, the adjoint problem), primarily in the space $L_2(0,1)$). Additionally, works [8-13] investigated similar problems in $L_p(0,1)$, and provided abstract approaches for their analysis. However, in most cases, the boundary forms generated an unbounded functional in the considered space. In this case, the operator had a dense domain, allowing the construction of an adjoint operator or assuming the regularity of the boundary conditions [1, 2, 4, 8-12]. Here, we will consider integral boundary conditions of the form

$$U_\nu (y) = \int_0^1 \varphi_\nu (x)\, y (x) dx = 0,\ \nu = 1, 2, \tag{3}$$

where $\varphi_\nu (x)$ - are given linearly independent functions belonging to the space $L_q (0, 1)$, $\frac{1}{p} + \frac{1}{q} = 1$. These conditions are not regular in the sense of Birkhoff [1], and there is no corresponding adjoint operator for them. Similar conditions have been used for other purposes in [6, 7]. In [14], the problem (1),(3) was studied under stricter conditions on the functions $q (x)$ and $\varphi_\nu (x)$, where the asymptotics of eigenvalues and eigenfunctions were obtained. In [15], a Riesz basis theorem for the system of eigenfunctions in a certain subspace of $L_2 (0, 1)$ was proven. In [16], the completeness and minimality of the eigenfunctions in a certain subspace of $L_p (0, 1)$ were proven. It is worth noting that differential equations with nonlocal integral-type conditions have interesting applications in mechanics [17] and in the theory of diffusion processes [18].

2. Preliminaries

We define the differential operator L in the space $L_p (0, 1)$, $1 < p < \infty$, with the domain

$$D(L) = \{y(x) \in W_p^2(0, 1);\ l(y) \in L_p(0, 1);\ U_1(y) = 0,\ U_2(y) = 0\},$$

which acts as

$$Ly = l(y),\ \forall y \in D(y).$$

Consider the eigenvalue problem for the operator L:

$$Ly = \lambda y. \tag{4}$$

Let us set $\lambda = \rho^2$ and denote by $y_1(x, \rho)$ the solution of equation (4) that satisfies the initial conditions $y(0) = 0,\ y'(0) = \rho$, and by $y_2(x, \rho)$– the solution of equation (4) that satisfies the initial conditions $y(0) = 1,\ y'(0) = 0$. It is obvious that these solutions are linearly independent (i.e., they form a fundamental system of solutions of equation (4)). It is known [2] that these functions are also solutions of the following integral equations:

$$y_1 (x, \rho) = \sin\rho x + \frac{1}{\rho}\int_0^x q (t)\, y_1 (t, \rho) \sin\rho (x - t)\ dt, \tag{5}$$

$$y_2 (x, \rho) = \cos\rho x + \frac{1}{\rho}\int_0^x q (t)\, y_2 (t, \rho) \sin\rho (x - t)\ dt. \tag{6}$$

These solutions satisfy the estimates as $| \rho | \to \infty$ (see. [2]):

$$| y_1 (x, \rho) | \le Ce^{|Im\rho| x},\ \ | y_2 (x, \rho) | \le Ce^{|Im\rho| x}\ , \tag{7}$$

and are therefore bounded in the strip $| Im\rho | \le h$, for some $h > 0$. For further purposes, we need to refine these estimates.

Lemma 1. *For the functions $y_1(x,\rho)$ and $y_2(x,\rho)$ in the strip $\mid Im\rho \mid\le h$, the following asymptotic formulas hold as $\mid \rho \mid\to\infty$:*

$$y_1(x,\rho)=\sin\rho x-\frac{1}{2\rho}\int_0^x q(t)\,dt\cos\rho x\ +\frac{1}{2\rho}\int_0^x q(t)\cos\rho(x-2t)\ dt+O\left(\frac{1}{\rho^2}\right); \quad (8)$$

$$y_2(x,\rho)=\cos\rho x\ +\frac{1}{2\rho}\int_0^x q(t)\,dt\ \sin\rho x\ +\frac{1}{2\rho}\int_0^x q(t)\sin\rho(x-2t)\ dt+O\left(\frac{1}{\rho^2}\right). \quad (9)$$

Proof. Substituting the expressions for the functions $y_1(t,\rho)$, $y_2(t,\rho)$, given by formulas (5) and (6), into the right-hand side of formulas (5) and (6) under the integral signs:

$$y_1(x,\rho)=\sin\rho x\ +\frac{1}{\rho}\int_0^x q(t)\left[\sin\rho t\ +\frac{1}{\rho}\int_0^t q(\tau)\,y_1(\tau,\rho)\sin\rho(t-\tau)\ d\tau\right]\sin\rho(x-t)\ dt=$$

$$=\sin\rho x\ +\frac{1}{\rho}\int_0^x q(t)\sin\rho t\ \ \sin\rho(x-t)\ dt+$$

$$+\frac{1}{\rho^2}\int_0^x q(t)\left(\int_0^t q(\tau)\,y_1(\tau,\rho)\sin\rho(t-\tau)\ d\tau\right)\sin\rho(x-t)\ dt=$$

$$=\sin\rho x\ -\frac{1}{2\rho}\int_0^x q(t)\,dt\cos\rho x\ +\frac{1}{2\rho}\int_0^x q(t)\cos\rho(x-2t)\ dt+\frac{1}{\rho^2}r_1(x,\rho);$$

$$y_2(x,\rho)=\cos\rho x\ +\frac{1}{\rho}\int_0^x q(t)\left[\cos\rho t\ +\frac{1}{\rho}\int_0^t q(\tau)\,y_2(\tau,\rho)\,d\tau\right]\sin\rho(x-t)\ dt=$$

$$=\cos\rho x\ +\frac{1}{\rho}\int_0^x q(t)\cos\rho t\ \sin\rho(x-t)\ dt+$$

$$+\frac{1}{\rho^2}\int_0^x q(t)\left(\int_0^t q(\tau)\,y_2(\tau,\rho)\sin\rho(t-\tau)\ d\tau\right)\sin\rho(x-t)\ dt=$$

$$=\cos\rho x\ +\frac{1}{2\rho}\int_0^x q(t)\,dt\ \sin\rho x\ +\frac{1}{2\rho}\int_0^x q(t)\sin\rho(x-2t)\ dt+\frac{1}{\rho^2}r_2(x,\rho),$$

where denoted by

$$r_i(x,\rho)=\int_0^x q(t)\left(\int_0^t q(\tau)\,y_i(\tau,\rho)\sin\rho(t-\tau)\ d\tau\right)\sin\rho(x-t)\ dt,\ i=1,2.$$

Considering the known inequalities

$$\mid\sin\rho x\mid\le e^{|Im\rho|x},\ \mid\cos\rho x\mid\le e^{|Im\rho|x},\ x\in[0,1],$$

As well as inequalities (7), we obtain

$$\mid r_i(x,\rho)\mid\le\int_0^x|q(t)|\left(\int_0^t|q(\tau)|\,|y_i(\tau,\rho)|\,|\sin\rho(t-\tau)\ |\,d\tau\right)|\sin\rho(x-t)\ |\,dt\le$$

$$\leq C\int_0^x |q(t)| \left(\int_0^t |q(\tau)|\, e^{|Im\rho|\tau} e^{|Im\rho|(t-\tau)} d\tau\right) e^{|Im\rho|(x-t)} dt \leq$$

$$\leq Ce^{|Im\rho|x}\int_0^x |q(t)| \left(\int_0^t |q(\tau)|\, d\tau\right) dt \leq Ce^{|Im\rho|}\left(\int_0^1 |q(t)|\, dt\right)^2.$$

Thus, for $|Im\rho| \leq h, \quad |\rho| \to \infty$, the following estimates hold uniformly for $x \in [0,1]$:

$$|r_1(x,\rho)| = O(1),\ |r_2(x,\rho)| = O(1),$$

From this, the validity of formulas (8) and (9). Lemma is proved.

Let us present briefly the main definitions and facts which will be used in what follows. Let X be a B-space. A system$\{x_n\}_{n\in N}$ of elements X is said to be complete in X if $\overline{L(\{x_n\}_{n\in N})} = X$; that is, any element of the space X can be approximated by a linear combination of elements of this system with any accuracy in the norm of the space X.

A system $\{x_n\}_{n\in N}$ of elements X is said to be minimal in X if $x_n \notin \overline{L(\{x_k\}_{k\neq n})}$. It is well known that a system $\{x_n\}_{n\in N}$ is minimal if and only if there exists a biorthogonal system which is dual to it, that is, a system of linear functionals $\{x_n^*\}_{n\in N}$ from X^* such that $\langle x_n, x_k^*\rangle = \delta_{nk}$ for all $n,\ k \in N$. Moreover, if the initial system is complete and minimal in X, then the biorthogonal system is uniquely defined.

A system $\{x_n\}_{n\in N}$ forms a basis of the space X if, for any element $x \in X$, there exists a unique expansion into a series

$$x = \sum_{n=1}^{\infty} c_n x_n$$

converging in the norm of the space X.

Two systems $\{x_n\}_{n\in N}$ and $\{y_n\}_{n\in N}$ of a B-space X are called equivalent if there exists an automorphism that maps one of these systems to the other. A system equivalent to a basis is itself a basis in the same space.

Any basis is a complete and minimal system in X, and, therefore, we can uniquely find its biorthogonal dual system $\{x_n^*\}_{n\in N}$ and hence the expansion of any element $x \in X$ with respect to the basis $\{x\}_{n\in N}$ coincides with its biorthogonal expansion, that is, $c_n = \langle x, x_n^*\rangle$ for all $n \in N$.

We will use also some facts about p - closure bases. Concerning these facts more details one can see the works [19, 20].

Systems $\{x_n\}_{n\in N}, \ \ \{y_n\}_{n\in N} \subset X$ in B -space X are called p-closure if

$$\sum_{n=1}^{\infty} \|x_n - y_n\|_X^p < \infty.$$

The minimal system $\{x_n\}_{n\in N} \subset X$ with biorthogonal system $\{x_n^*\}_{n\in N} \subset X^*$ is called p- besselian, if for any $x \in X$

$$\left(\sum_{n=1}^{\infty} |\langle x, x_n^*\rangle|^p\right)^{\frac{1}{p}} \leq M\|x\|_X.$$

If the basis $\{x_n\}_{n\in N}$ for X is p- besselian, then we call it as p-basis.

It is valid the following

Theorem 1. *[20] Let the system $\{x_n\}_{n\in N}$ is p -basis for B- space X and the system $\{y_n\}_{n\in N}\subset X$ is p'– clouser to it, $1<p<\infty$. Then the following assertions are equivalent:*

1. *$\{y_n\}_{n\in N}$ is complete in X;*
2. *$\{y_n\}_{n\in N}$ is minimal in X;*
3. *$\{y_n\}_{n\in N}$ is isomorphic to $\{x_n\}_{n\in N}$ basis for X.*

3. Asymptotics of Eigenvalues and Eigenfunctions

The main result of this section is:

Theorem 2. *Let the function $q(x)$ be integrable, and k, let the functions $\varphi_\nu(x), \nu=1,2,$ belong to the class $W_1^2(0,1)$. Suppose that*

$$\alpha_1\beta_2-\alpha_2\beta_1\neq 0, \tag{10}$$

where we denote $\alpha_\nu=\varphi_\nu(0)$, $\beta_\nu=\varphi_\nu(1)$. Then the eigenvalues of the operator L are asymptotically simple, and the following asymptotic formula holds:

$$\lambda_k=\rho_k^2,\quad \rho_k=\pi k+O\left(\frac{1}{k}\right),\quad k\to\infty.$$

Proof. To find the eigenvalues of the operator L consider the determinant

$$\Delta(\rho)=\begin{vmatrix} U_1(y_1) & U_1(y_2)\\ U_2(y_1) & U_2(y_2)\end{vmatrix},$$

where $y_1=y_1(x,\rho)$, $y_2=y_2(x,\rho)$ – form a fundamental system of solutions of equation (4). Using formulas (8) and (9), we obtain

$$U_\nu(y_1)=\int_0^1 y_1(x,\rho)\varphi_\nu(x)\,dx=\int_0^1\varphi_\nu(x)\sin\rho x\,dx-\frac{1}{2\rho}\int_0^1\varphi_\nu(x)\cos\rho x\int_0^x q(t)\,dtdx+$$

$$+\frac{1}{2\rho}\int_0^1\varphi_\nu(x)\int_0^x q(t)\cos\rho(x-2t)\,dtdx+O(\frac{1}{\rho^2})=$$

$$=\frac{1}{\rho}\left(-\varphi_\nu(x)\cos\rho x+\varphi'_\nu(x)\sin\rho x\right)\Big|_0^1-\frac{1}{\rho^2}\int_0^1\varphi''_\nu(x)\sin\rho x\,dx-$$

$$+\frac{1}{2\rho}\int_0^1 q(t)\left(\int_t^1\varphi_\nu(x)\,cos\rho(x-2t)\,dx\right)dt+O\left(\frac{1}{\rho^2}\right)=$$

$$=\frac{1}{\rho}\left(-\beta_\nu\cos\rho+\alpha_\nu\right)+\frac{1}{\rho^2}\varphi'_\nu(1)\sin\rho-\frac{1}{\rho^2}\int_0^1\varphi"_\nu(x)\sin\rho\, xdx-$$

$$-\frac{1}{2\rho^2}\varphi_\nu(1)\sin\rho\ \int_0^1 q(t)\,dt+\frac{1}{2\rho^2}\int_0^1\varphi'_\nu(x)\int_0^x q(t)\,dt\ \sin\rho x\ dx+$$

$$+\frac{1}{2\rho^2}\int_0^1\varphi_\nu(x)\,q(x)\sin\rho x\ dx+\frac{1}{2\rho^2}\int_0^1 q(t)\left(\varphi_v(x)\sin\rho(x-2t)\ |_{x=t}^{x=1}\right)dt-$$

$$-\frac{1}{2\rho^2}\int_0^1 q(t)\int_t^1\varphi'_\nu(x)\sin\rho(x-2t)\ dxdt+O\left(\frac{1}{\rho^2}\right)=$$

$$=\frac{1}{\rho}\left(-\beta_\nu\cos\rho\ +\alpha_\nu\right)+O\left(\frac{1}{\rho^2}\right); \tag{11}$$

$$U_\nu(y_2)=\int_0^1 y_2(x,\rho)\varphi_\nu(x)\,dx=\int_0^1\varphi_\nu(x)\cos\rho x\ dx+\frac{1}{2\rho}\int_0^1\varphi_\nu(x)\sin\rho x\ \int_0^x q(t)\,dtdx+$$

$$+\frac{1}{2\rho}\int_0^1\varphi_\nu(x)\int_0^x q(t)\sin\rho(x-2t)\ dtdx+O(\frac{1}{\rho^2})=$$

$$=\frac{1}{\rho}\beta_\nu\sin\rho\ -\frac{1}{\rho^2}\varphi'_\nu(1)\cos\rho\ -\frac{1}{\rho^2}\int_0^1\varphi''_\nu(x)\cos\rho x\ dx-$$

$$-\frac{1}{2\rho^2}\left(\varphi_\nu(x)\int_0^x q(t)\,dt\cos\rho x\right)\Bigg|_{x=0}^{x=1}+\frac{1}{2\rho^2}\int_0^1 q(t)\int_t^1\varphi_v(x)\sin\rho(x-2t)\ dtdx=$$

$$=\frac{1}{\rho}\beta_\nu\sin\rho\ -\frac{1}{\rho^2}\varphi'_\nu(1)\cos\rho\ -\frac{1}{\rho^2}\int_0^1\varphi''_\nu(x)\cos\rho\ xdx-\varphi_\nu(1)\cos\rho\ \int_0^1 q(t)\,dt-$$

$$-\frac{1}{2\rho^2}\int_0^1 q(t)\left(\varphi_\nu(x)\ \cos\rho(x-2\mathrm{t})\ |_{x=t}^{x=1}-\int_t^1\varphi'_\nu(x)\cos\rho(x-2t)\ dx\right)dt+O\left(\frac{1}{\rho^2}\right)=$$

$$=\frac{1}{\rho}\beta_\nu\sin\rho\ +O\left(\frac{1}{\rho^2}\right). \tag{12}$$

Thus, the following relations are obtained:

$$U_\nu(y_1)=\frac{1}{\rho}\left(-\beta_\nu\cos\rho\ +\alpha_\nu\right)+O\left(\frac{1}{\rho^2}\right),$$

$$U_\nu(y_2)=\frac{1}{\rho}\beta_\nu\sin\rho\ +O\left(\frac{1}{\rho^2}\right).$$

Taking these relations into account, we obtain

$$\varDelta(\rho)=\begin{vmatrix}\frac{1}{\rho}\left(-\beta_1\cos\rho\ +\alpha_1\right)+O\left(\frac{1}{\rho^2}\right) & \frac{1}{\rho}\beta_1\sin\rho\ +O\left(\frac{1}{\rho^2}\right)\\ \frac{1}{\rho}\left(-\beta_2\cos\rho\ +\alpha_2\right)+O\left(\frac{1}{\rho^2}\right) & \frac{1}{\rho}\beta_2\sin\rho\ +O\left(\frac{1}{\rho^2}\right)\end{vmatrix}=$$

$$=\frac{1}{\rho^2}\begin{vmatrix}-\beta_1\cos\rho\ +\alpha_1+O\left(\frac{1}{\rho}\right) & \beta_1\sin\rho\ +O\left(\frac{1}{\rho}\right)\\ -\beta_2\cos\rho\ +\alpha_2+O\left(\frac{1}{\rho}\right) & \beta_2\sin\rho\ +O\left(\frac{1}{\rho}\right)\end{vmatrix}=$$

$$= \frac{1}{\rho^2}\left(\alpha_1\beta_2 - \alpha_2\beta_1\right)\sin\rho \ + O\left(\frac{1}{\rho^3}\right) = \frac{1}{\rho^2}\left(\left(\alpha_1\beta_2 - \alpha_2\beta_1\right)\sin\rho \ + O\left(\frac{1}{\rho}\right)\right).$$

According to condition (10) $\alpha_1\beta_2 - \alpha_2\beta_1 \neq 0$, applying Rouché's theorem and using the standard method (see [1, p. 77]), we obtain the asymptotics of the zeros of $\Delta(\rho)$:

$$\rho_k = \pi k + O\left(\frac{1}{k}\right), \quad k \to \infty.$$

The theorem is proved.

Remark 1. *The numbering of the zeros will be refined later. For now, we can only assert that they are asymptotically simple. This means that the number of associated functions of the operator L (if any exist) is finite.*

Now, let us proceed to finding the eigenfunctions of the operator L. The following theorem holds:

Theorem 3. *Under the conditions of Theorem 2, the eigenfunctions of the operator L satisfy the following asymptotic formula:*

$$y_k(x) = \cos\pi kx \ + O\left(\frac{1}{k}\right), \ k \to \infty. \tag{13}$$

Proof. Following [1, p. 84], we seek the eigenfunctions of the operator L in the form

$$y_k(x) = c_k \begin{vmatrix} y_1(x,\rho_k) & y_2(x,\rho_k) \\ U_2(y_1) & U_2(y_2) \end{vmatrix}, \tag{14}$$

if $\alpha_2 \neq \pm\beta_2$, or

$$y_k(x) = c_k \begin{vmatrix} U_1(y_1) & U_1(y_2) \\ y_1(x,\rho_k) & y_2(x,\rho_k) \end{vmatrix}, \tag{15}$$

if $\alpha_1 \neq \pm\beta_1$, where c_k are some normalization factors to be determined. Assume $\alpha_2 \neq \pm\beta_2$. We have $\rho_k = \pi k + O\left(\frac{1}{k}\right)$ and

$$y_1(x,\rho_k) = \sin\rho_k x \ + O\left(\frac{1}{\rho_k}\right) = \sin\pi kx + O\left(\frac{1}{k}\right),$$

$$y_2(x,\rho_k) = \cos\rho_k x \ + O\left(\frac{1}{\rho_k}\right) = \cos\pi kx + O\left(\frac{1}{k}\right).$$

Substituting these expressions into determinant (14) and using formulas (11) and (12) for $\rho = \rho_k$, we obtain

$$y_k(x) = c_k \begin{vmatrix} \sin\pi kx + O\left(\frac{1}{k}\right) & \cos\pi kx + O\left(\frac{1}{k}\right) \\ \frac{1}{\rho_k}\left(-\beta_2\cos\rho_k \ + \alpha_2\right) + O\left(\frac{1}{\rho_k^2}\right) & \frac{1}{\rho}\beta_2\sin\rho_k \ + O\left(\frac{1}{\rho_k^2}\right) \end{vmatrix} =$$

$$c_k \begin{vmatrix} \sin\pi kx + O\left(\frac{1}{k}\right) & \cos\pi kx + O\left(\frac{1}{k}\right) \\ \frac{1}{\pi k}\left(-\beta_2(-1)^k + \alpha_2\right) + O\left(\frac{1}{k^2}\right) & \frac{1}{\rho}\beta_2 \sin\rho_k \ + O\left(\frac{1}{k^2}\right) \end{vmatrix} =$$

$$= c_k \left(\frac{1}{\pi k}\left(\alpha_2 - (-1)^k \beta_2\right)\cos\pi kx \ + O\left(\frac{1}{k^2}\right)\right).$$

By choosing $c_k = \frac{\pi k}{\left(\alpha_2 - (-1)^k \beta_2\right)}$, from the last equation, we obtain the validity of (13).
If $\alpha_2 = \pm\beta_2$, then from condition (10), it follows that $\alpha_1 \neq \pm\beta_1$. Proceeding similarly to the previous case and using formulas (11), (12), and the asymptotics of ρ_k, from formula (15) we obtain the validity of (13), where we should choose $c_k = \frac{\pi k}{\alpha_1 - (-1)^k \beta_1}$.

The theorem is proved.

4. Basis Property of Eigenfunctions

The operator L constructed in Section 2 does not have a dense domain in $L_p(0,1)$, and therefore the system of eigenfunctions and associated functions of this operator cannot be complete, let alone form a basis in this space. To address this issue, we consider the operator L not in the entire space $L_p(0,1)$, but in its closed subspace:

$$X_p = \{f(x) \in L_p(0,1) : U_\nu(f) = 0, \ \nu = 1,2\}.$$

Since the functionals U_ν, $\nu = 1,2$, are bounded in $L_p(0,1)$, it is clear that codim $X_p = 2$.
We define the operator L in the space X_p as follows: $D(L) = \left\{y \in W_p^2(0,1) \cap X_p : \ l(y) \in X_p \ \right\}$ and for $y \in D(L): \ Ly = l(y)$.
It is evident that the operator defined in this way acts in the space X_p and has a dense domain in this space.
The following theorem was proved in [16].

Theorem 4. *[16] The eigenfunctions and associated functions of the operator L form a complete and minimal system in the space* X_p, $1 \leq p < \infty$.

The main result of this work is the following theorem.

Theorem 5. *There exist functions* $\psi_k \in L_q(0,1)$, $k = 1,2$, *such that the system* $\{\psi_1, \psi_2\} \cup \{y_n\}_{n \geq 2}$ *forms a basis in the space* $L_p(0,1)$, $1 < p < \infty$, *equivalent to the system* $\{\cos\pi nx\}_{n=0}^{\infty}$. *The system* $\{y_n\}_{n\geq 2}$ *of eigenfunctions and associated functions of the operator L forms a basis in the space* X_p, $1 < p < \infty$.

Proof. Since codim $X_p = 2$, the space $L_p(0,1)$ can be represented as a direct sum: $L_p(0,1) = X_p \oplus X_0$, where dim $X_0 = 2$. According to the Hahn-Banach theorem, there exist functions $\psi_k \in L_q(0,1)$, $k = 1,2$, such that $\langle\varphi_i, \psi_k\rangle = \delta_{ik}$, $i,k = 1,2$, and $\langle y, \psi_k\rangle = 0$, $\forall y \in X_p$, where φ_1, $\varphi_2-$are functions from the boundary conditions.

From Theorem 4, it follows that the system $\{\psi_1, \psi_2\} \cup \{y_n\}$ is complete and minimal in $L_p(0,1)$. Indeed, let $\{z_n\}$ be the system of eigenfunctions and associated functions of the adjoint operator L^*, forming a biorthogonal system with $\{y_n\}$. Then the system $\{\varphi_1, \varphi_2\} \cup \{z_n\}$ is biorthogonal to $\{\psi_1, \psi_2\} \cup \{y_n\}$, which is equivalent to the minimality of $\{\psi_1, \psi_2\} \cup \{y_n\}$ in $L_p(0,1)$. Now, we prove the completeness of this system. Suppose $z \in L_q(0,1)$, $\frac{1}{p} + \frac{1}{q} = 1$, and $\langle \psi_1, z\rangle = \langle \psi_2, z\rangle = \langle y_n, z\rangle = 0$. For any $\forall f \in L_p(0,1)$ we represent it as $f = y + c_1\psi_1 + c_2\psi_2$, where $y \in X_p$. Then $\langle f, z\rangle = \langle y, z\rangle + c_1 \langle \psi_1, z\rangle + c_2 \langle \psi_2, z\rangle = \langle y, z\rangle$. On the other hand, since the system $\{y_n\}$ is complete in X_p, and $\langle y_n, z\rangle = 0$, $\forall n$, it follows that $\langle y, z\rangle = 0$, $\forall y \in X_p$. Thus, we obtain that $\langle f, z\rangle = 0$, $\forall f \in L_p(0,1)$, which implies $z = 0$, meaning that the system $\{\psi_1, \psi_2\} \cup \{y_n\}$ is complete in $L_p(0,1)$. Now we show that this system forms a basis in $L_p(0,1)$. From Theorem 3, it follows that the systems $\{y_n\}$ and $\{\cos\pi nx\}$ are $s-$close for any $s \in (1,\infty)$:

$$\sum_{n=0}^{\infty} \|y_n(x) - \cos\pi nx\|_{L_p}^{s} < +\infty, \tag{16}$$

where $y_0(x) = \psi_1(x), y_1(x) = \psi_2(x)$. Let $p \in (1,2]$, then by the Hausdorff-Young inequality (see [21]), for all $\forall f \in L_p(0,1)$:

$$\left(\sum_{n=0}^{\infty} |\langle f, e_n\rangle|^q\right)^{\frac{1}{q}} \leq C\|f\|_{L_p}, \tag{17}$$

where $e_n(x) = 2\cos\pi nx$. This inequality implies that the system $\{\cos\pi nx\}_{n=0}^{\infty}$ forms a $q-$basis in $L_p(0,1)$. For $p \in (2,\infty)$, we have $q \in (1,2)$ and the continuous embedding $L_p(0,1) \subset L_q(0,1)$ holds. Again, using the Hausdorff-Young inequality: $\forall f \in L_p(0,1)$:

$$\left(\sum_{n=0}^{\infty} |\langle f, e_n\rangle|^p\right)^{\frac{1}{p}} \leq C\|f\|_{L_q} \leq C\|f\|_{L_q}, \tag{18}$$

that is, the system $\{\cos\pi nx\}_{n=0}^{\infty}$ forms $p-$basis in $L_p(0,1)$. Thus, denoting $r = \max\{p,q\}$ and choosing in inequality (16) $s = r(r-1)^{-1}$, and taking into account inequalities (17) and (18), we obtain that for any $p \in (1,\infty)$, all the conditions of Theorem 2.1 are satisfied, according to which the system $\{y_n(x)\}_{n=0}^{\infty}$ forms a basis in $L_p(0,1)$, equivalent to the system $\{\cos\pi nx\}_{n=0}^{\infty}$. Moreover, this result, as well as the asymptotics of the eigenfunctions, dictates that the numbering of the eigenfunctions and associated functions of the operator L should be performed as $\{y_n(x)\}_{n=2}^{\infty}$. Now, it is easy to establish the basis property of the system in the space X_p. Let $y(x)$ be an arbitrary function from X_p. Since in this case $\langle y, \varphi_1\rangle = \langle y, \varphi_2\rangle = 0$, its expansion in the basis $\{y_n(x)\}_{n=0}^{\infty}$ in $L_p(0,1)$ takes the form:

$$y = \sum_{n=2}^{\infty} \langle y, z_n\rangle y_n.$$

Furthermore, from the completeness and minimality of the system $\{y_n(x)\}_{n=2}^{\infty}$ in X_p we conclude that such an expansion is unique, i.e., the system $\{y_n(x)\}_{n=2}^{\infty}$ forms a basis in X_p.

Corollary 1. *The system $\{y_n(x)\}_{n=2}^{\infty}$ is an $r-$ basis in the space X_p, $1 < p < \infty$, where $r = \max\{p,q\}$.*

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